



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No. 1

1. Experiment Title: Demonstrate Inline, Internal, External JavaScript, use console methods; create a webpage showing user info and a welcome message.

2. Software/Tools Required: VS Code, Html/JS Compiler...

3. Experiment Program Code:

Internal:

```
<!DOCTYPE html>

<html>

<head>

    <title>Hello World</title>


<style>

    body{

        background-color:#ffe4e1;

        text-align:center;

        font-family:Arial, sans-serif;

        padding:40px;

    }


    h1{

        font-size:48px;

        color:white;

        background-color:#91ff00;

        padding:15px;

        border-radius:8px;

    }


    p{
```



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```
font-size:20px;

color:#333333;

background-color:#ffffcc;

padding:10px;

margin-top:20px;

}

button{

font-size:18px;

color:white;

background-color:green;

padding:12px 24px;

border:none;

border-radius:5px;

cursor:pointer;

margin-top:20px;

}

</style>

</head>

<body>

<h1>Hello World</h1>

<p>Name : Aaryan K. Kawale , PRN : 24070521162</p>

<p>This is a sample Webpage made using External css.</p>

<button>Click Me</button>

</body>

</html>
```



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External:

```
<!DOCTYPE html>

<html>

<head>

  <title>Hello World</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>


  <h1>Hello World</h1>

  <p> Name : Aaryan Kawale , PRN: 24070521162</p>

  <p> Sec: C, Sem: 5th </p>


  <p>This is a sample Webpage is made by using the External CSS.</p>


  <button>Click Me</button>


</body>

</html>
```

Inline:

```
<!DOCTYPE html>

<html>

<head>

  <title>Hello World</title>

</head>

<body style="background-color:#f0f8ff; text-align:center; font-family:Arial, sans-serif;
padding:40px;">
```



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```
<h1 style="font-size:48px; color:white; background-color:#007BFF; padding:15px; border-radius:8px;">
```

Hello World

```
</h1>
```

```
<p>
```

Name : Aaryan K. Kawale ,PRN : 24070521162

```
</p>
```

```
<p style="font-size:20px; color:#333333; background-color:#ffffcc; padding:10px; margin-top:20px;">
```

This is a sample webpage designed by Inline CSS.

```
</p>
```

```
<button style="font-size:18px; color:white; background-color:green; padding:12px 24px; border:none; border-radius:5px; cursor:pointer; margin-top:20px;">
```

Click Me

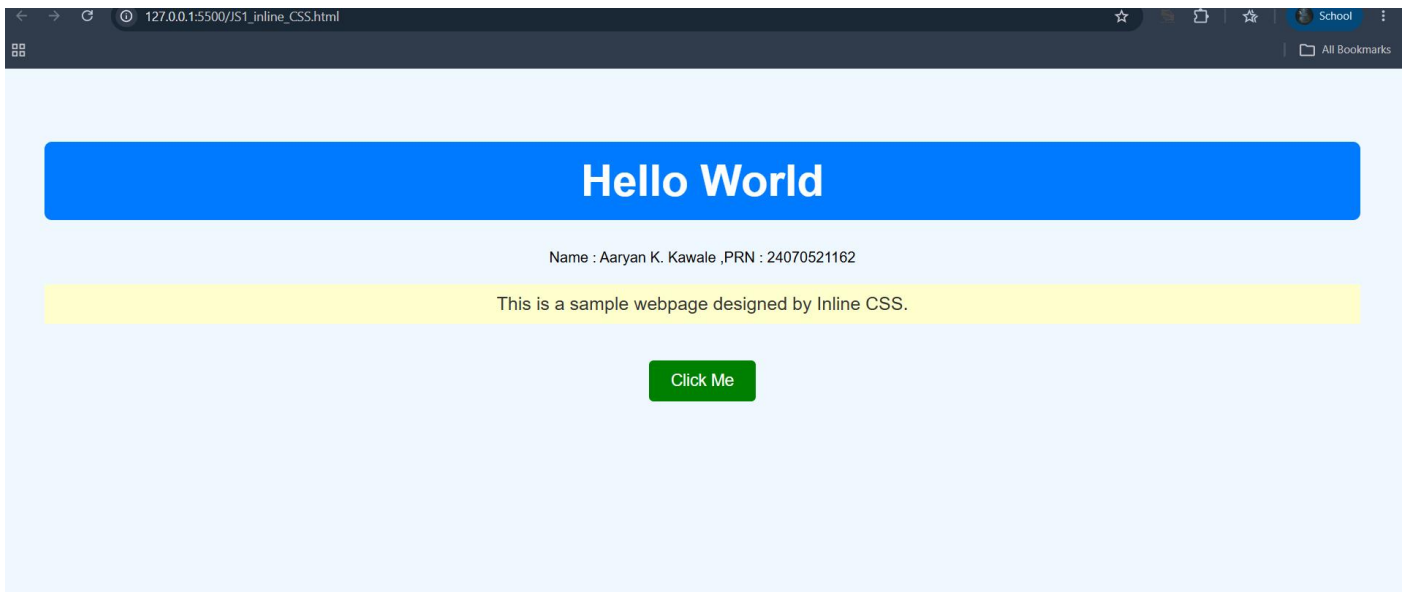
```
</button>
```

```
</body>
```

```
</html>
```

4. Output:

Inline:





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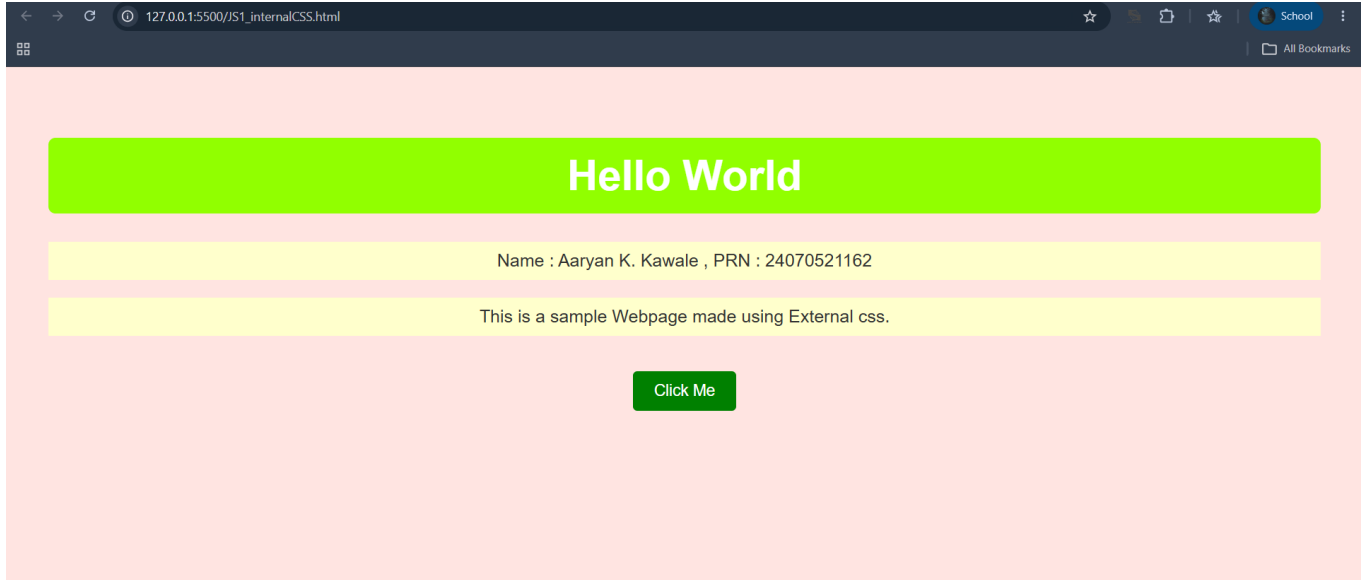
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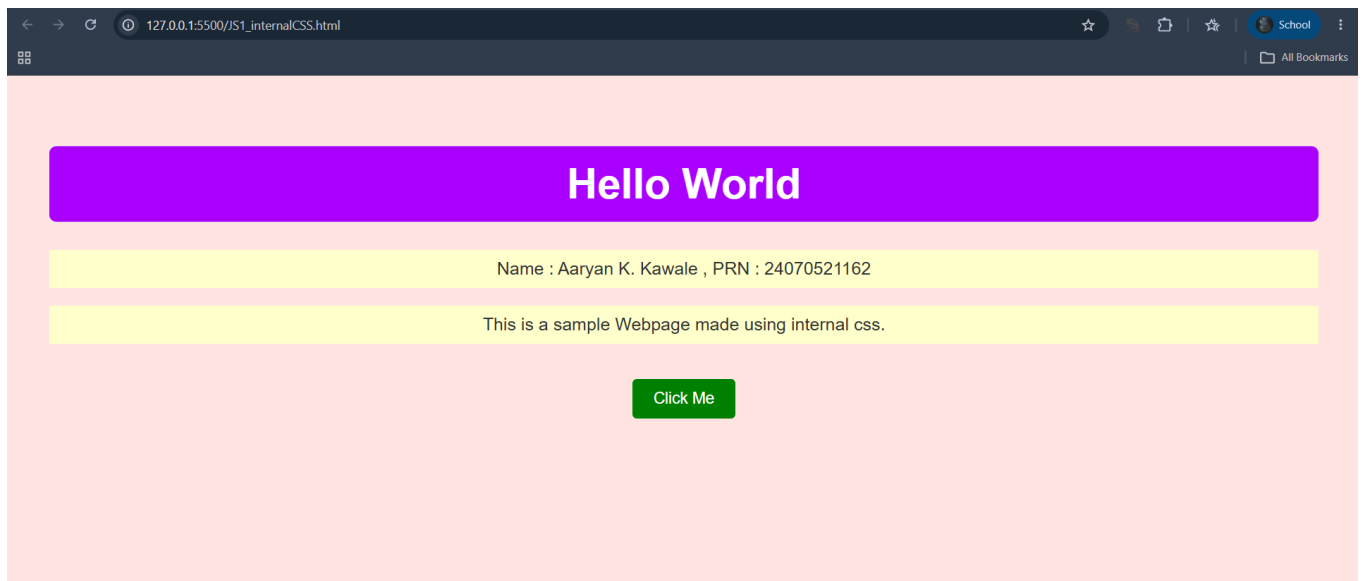
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External:



Internal:





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Case Study 1

1. **Case Study Title:** Desing a webpage for SIT Nagpur with buttons , text , images, etc...

2. **Case Study Program Code:**

```
3. <!DOCTYPE html>
4. <html lang="en">
5. <head>
6.     <meta charset="UTF-8">
7.     <meta name="viewport" content="width=device-width, initial-
scale=1.0">
8.     <title>SIT Nagpur - Computer Science Department</title>
9.     <style>
10.         body {
11.             font-family: Arial, sans-serif;
12.             margin: 0;
13.             padding: 0;
14.             /* Live campus background image */
15.             background-image:
url('https://d3lzc6mbbadaf.cloudfront.net/media/details/ANI-
20230411114128.jpg');
16.             background-size: cover;
17.             background-position: center;
18.             background-attachment: fixed;
19.             background-repeat: no-repeat;
20.         }
21.         header {
22.             background-color: #ae1717b8; /* Symbiosis Maroon */
23.             color: white;
24.             padding: 20px;
25.             text-align: center;
26.         }
27.         .logo {
28.             max-width: 180px;
29.             height: auto;
30.             margin-bottom: 10px;
31.             background-color: transparent;
32.         }
33.         nav {
34.             background-color: #333;
35.             overflow: hidden;
36.         }
37.         nav a {
38.             float: left;
39.             display: block;
40.             color: white;
41.             text-align: center;
```



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```
42.         padding: 14px 20px;
43.         text-decoration: none;
44.     }
45.     nav a:hover {
46.         background-color: #ddd;
47.         color: black;
48.     }
49.
50.     /* Flexbox wrapper to place sections side by side */
51.     .content-wrapper {
52.         display: flex;
53.         flex-direction: row;
54.         justify-content: center;
55.         align-items: stretch;
56.         gap: 30px;
57.         max-width: 1200px;
58.         margin: 50px auto 100px auto;
59.         padding: 0 20px;
60.     }
61.
62.     /* Shared stylings for both white info blocks */
63.     .info-box {
64.         flex: 1;
65.         padding: 30px;
66.         background: rgba(255, 255, 255, 0.92);
67.         border-radius: 8px;
68.         box-shadow: 0 4px 15px rgba(0,0,0,0.3);
69.     }
70.
71.     h2, h3 {
72.         color: #333;
73.     }
74.     ul {
75.         padding-left: 20px;
76.         line-height: 1.6;
77.     }
78.     li {
79.         margin-bottom: 8px;
80.     }
81.
82.     footer {
83.         text-align: center;
84.         padding: 15px;
85.         background-color: #333;
86.         color: white;
87.         position: fixed;
88.         bottom: 0;
89.         width: 100%;
```



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```
90.     }
91.
92.     /* Responsive design for tablets and mobile viewports */
93.     @media (max-width: 768px) {
94.         .content-wrapper {
95.             flex-direction: column;
96.             margin-top: 20px;
97.         }
98.     }
99. </style>
100. </head>
101. <body>
102.
103.     <header>
104.         
107.         <h1>Department of Computer Science & Engineering</h1>
108.         <p>Symbiosis Institute of Technology, Nagpur</p>
109.     </header>
110.
111.     <nav>
112.         <a href="https://sitnagpur.edu.in/">Home</a>
113.         <a href="student_info.html">Student Information</a>
114.     </nav>
115.
116.     <div class="content-wrapper">
117.
118.         <div class="info-box">
119.             <h2>Welcome to the Department</h2>
120.             <p>Welcome to the Department of CSE at SIT Nagpur.
121.             We are committed to delivering world-class engineering education,
122.             fostering innovation, and preparing our students to lead in the global
123.             technology landscape.</p>
124.
125.             <h3>Key Academic Highlights</h3>
126.             <ul>
127.                 <li>State-of-the-art Research & Computing
128.                 Labs</li>
129.                 <li>Industry-aligned & Integrated
130.                 Curriculum</li>
131.                 <li>Excellent Placement Support & Recruitment
132.                 Cells</li>
133.             </ul>
134.         </div>
135.
136.         <div class="info-box">
137.             <h2>Campus Infrastructure</h2>
```



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```
130.         <p>SIT Nagpur provides premium, modern amenities
              designed to enhance student living, academic research, and holistic
              wellness across the campus footprint.</p>
131.
132.         <h3>Facilities Provided</h3>
133.         <ul>
134.             <li><strong>Academic Blocks:</strong> Smart,
                  fully air-conditioned digital classrooms.</li>
135.             <li><strong>Central Library:</strong> Digital
                  library ecosystem packed with global e-resources.</li>
136.             <li><strong>Living Amenities:</strong> Wi-Fi
                  enabled secure hostels for girls & boys.</li>
137.             <li><strong>Dining Options:</strong> In-house
                  food mess alongside the modern "Symbi-Eats" cafeteria.</li>
138.             <li><strong>Sports & Gym:</strong> High-end
                  fitness setups, sports complex, and wide athletic grounds.</li>
139.         </ul>
140.     </div>
141.
142. </div>
143.
144.     <footer>
145.         &copy; 2026 Symbiosis Institute of Technology, Nagpur.
          All Rights Reserved.
146.     </footer>
147.
148.     <script src="script.js"></script>
149. </body>
150. </html>
```



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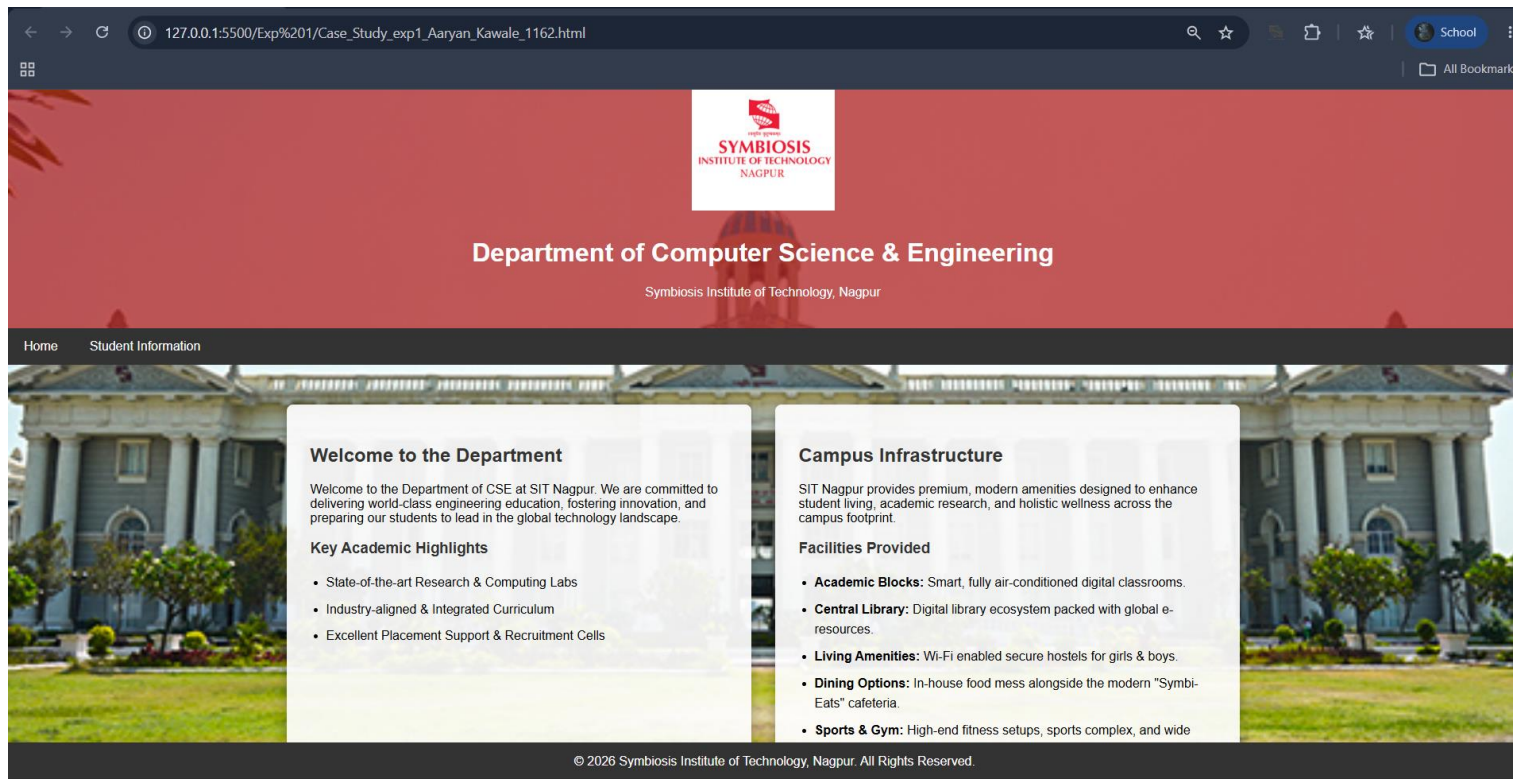
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3.Output:





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Experiment No. 2

- 1. Experiment Title:** Use var,let,const and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.
- 2. Software/Tools Required:** VS Code, Html/JS Compiler...
- 3. Experiment Program Code:**

```
4. <!DOCTYPE html>
5. <html>
6. <head>
7.     <title>Bill Receipt</title>
8.     <style>
9.         body {
10.             background-color: #f4f4f9;
11.             font-family: Arial, sans-serif;
12.             display: flex;
13.             justify-content: center;
14.             align-items: center;
15.             height: 100vh;
16.             margin: 0;
17.         }
18.         .receipt-box {
19.             background-color: #ffffff;
20.             border: 2px solid #333333;
21.             padding: 30px;
22.             width: 350px;
23.             box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
24.         }
25.         h3 {
26.             text-align: center;
27.             margin-top: 0;
28.         }
29.     </style>
30.</head>
31.<body>
32.
33.     <script>
34.         // 1. Popups to take user input
35.         var uName = prompt("Enter User Name:");
36.         var contact = prompt("Enter Contact No:");
37.         var pName = prompt("Enter Product Name:");
38.         var qty = Number(prompt("Enter Quantity:"));
39.         var price = Number(prompt("Enter Price per Unit:"));
```



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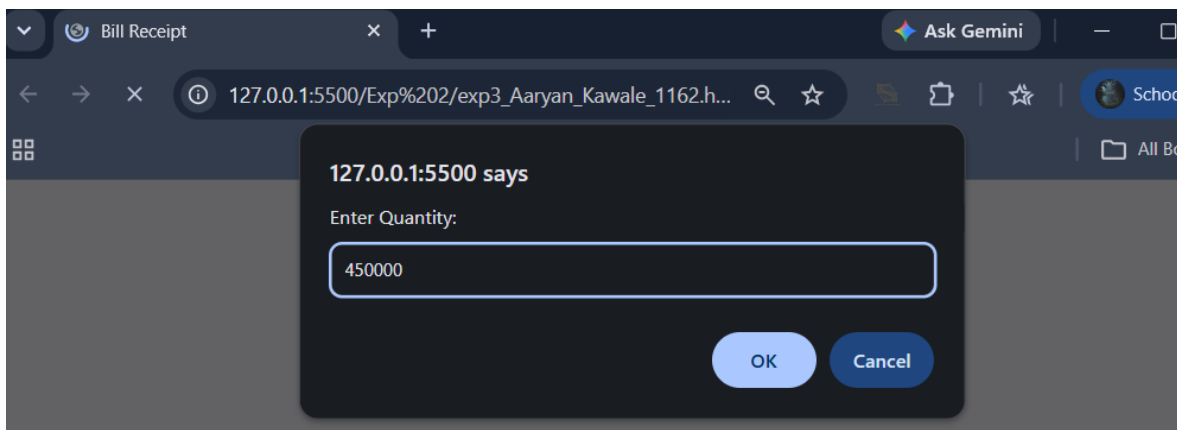
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```
40.
41.     // 2. Mathematical calculations
42.     var total = qty * price;
43.     var gst = total * 0.18;
44.     var finalAmount = total + gst;
45.
46.     // 3. Outputting only the centered bill box
47.     document.write("<div class='receipt-box'>");
48.     document.write("<h3>Bill Receipt</h3>");
49.     document.write("<p><b>User Name:</b> " + uName + "</p>");
50.     document.write("<p><b>Contact No:</b> " + contact + "</p>");
51.     document.write("<p><b>Product Name:</b> " + pName + "</p>");
52.     document.write("<p><b>Quantity:</b> " + qty + "</p>");
53.     document.write("<p><b>Price:</b> " + price + "</p>");
54.     document.write("<hr>");
55.     document.write("<p><b>Total Value:</b> " + total + "</p>");
56.     document.write("<p><b>GST (18%):</b> " + gst + "</p>");
57.     document.write("<p style='font-size: 18px;'><b>Final
Amount:</b> " + finalAmount + "</p>");
58.     document.write("<br>");
59.     document.write("<p style='font-size: 12px; color: #555; text-
align: center;'>Name: Aaryan Kawale, PRN: 24070521162</p>");
60.     document.write("</div>");
61. </script>
62.
63.</body>
64.</html>
```

4.Output:





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Bill Receipt

127.0.0.1:5500 says

Enter Contact No:

24070521162

OK Cancel

127.0.0.1:5500 says

Enter User Name:

Aaryan Kawale

OK Cancel

Bill Receipt

User Name: Aaryan Kawale

Contact No: 24070521162

Product Name: Asus

Quantity: 5

Price: 40000

Total Value: 200000

GST (18%): 36000

Final Amount: 236000

Name: Aaryan Kawale, PRN: 24070521162



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Case Study 2

1. **Case Study Title:** Develop the webpage for an online shopping website using html, css js .

2. Case Study Program Code:

```
3. <!DOCTYPE html>
4. <html lang="en">
5. <head>
6.   <meta charset="UTF-8">
7.   <meta name="viewport" content="width=device-width, initial-
  scale=1.0">
8.   <title>Simple Shop</title>
9.   <style>
10.    * { box-sizing: border-box; margin: 0; padding: 0; font-family:
  sans-serif; }
11.
12.    /* Split layout: Left column for items, right column for cart */
13.    body { display: flex; min-height: 100vh; background: #f4f4f9; }
14.
15.    .products-section { flex: 2; padding: 20px; }
16.    .cart-section { flex: 1; padding: 20px; background: #ffffff;
  border-left: 2px solid #ddd; }
17.
18.    h2 { margin-bottom: 15px; color: #333; }
19.
20.    /* Product Grid */
21.    .grid { display: grid; grid-template-columns: repeat(auto-fill,
  minmax(180px, 1fr)); gap: 15px; }
22.    .card { background: #fff; padding: 15px; border-radius: 8px;
  border: 1px solid #ddd; text-align: center; }
23.    .card h3 { font-size: 1.1rem; margin-bottom: 5px; }
24.    .price { color: #2563eb; font-weight: bold; margin-bottom: 10px; }
25.
26.    button { background: #2563eb; color: white; border: none; padding:
  8px 12px; border-radius: 5px; cursor: pointer; }
27.    button:hover { background: #1d4ed8; }
28.
29.    /* Cart Table & Summary */
30.    table { width: 100%; border-collapse: collapse; margin-bottom:
  20px; }
31.    th, td { text-align: left; padding: 8px; border-bottom: 1px solid
  #eee; }
32.    .remove-btn { background: #ef4444; padding: 4px 8px; font-size:
  0.8rem; }
```



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```
33.     .remove-btn:hover { background: #dc2626; }
34.
35.     .total-box { font-size: 1.2rem; font-weight: bold; display: flex;
36.     justify-content: space-between; margin-top: 15px; }
37.     .checkout-btn { width: 100%; padding: 10px; background: #10b981;
38.     margin-top: 15px; font-size: 1rem; }
39.     .checkout-btn:hover { background: #059669; }
40. </style>
41. </head>
42. <body>
43.     <!-- LEFT COLUMN: ITEMS -->
44.     <section class="products-section">
45.         <h2>Available Items</h2>
46.         <div class="grid" id="productGrid"></div>
47.     </section>
48.     <!-- RIGHT COLUMN: CART & TOTAL -->
49.     <section class="cart-section">
50.         <h2>Shopping Cart</h2>
51.         <table>
52.             <thead>
53.                 <tr>
54.                     <th>Item</th>
55.                     <th>Qty</th>
56.                     <th>Price</th>
57.                 </tr>
58.             </thead>
59.             <tbody id="cartTable"></tbody>
60.         </table>
61.
62.         <div class="total-box">
63.             <span>Total:</span>
64.             <span id="grandTotal">₹0.00</span>
65.         </div>
66.
67.         <button class="checkout-btn"
68.         onclick="checkout()">Checkout</button>
69.     </section>
70.
71.     <script>
72.         // 1. Items Data (Prices in INR)
73.         const products = [
74.             { id: 1, name: "Wireless Mouse", price: 1428.00 },
75.             { id: 2, name: "Keyboard", price: 2856.00 },
76.             { id: 3, name: "Headphones", price: 4284.00 },
77.             { id: 4, name: "USB Cable", price: 761.60 }
```



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```
78.     ];
79.
80.     let cart = [];
81.
82.     // 2. Render Product Cards (Left Column)
83.     function renderProducts() {
84.         const grid = document.getElementById("productGrid");
85.         grid.innerHTML = products.map(p => `
86.             <div class="card">
87.                 <h3>${p.name}</h3>
88.                 <p class="price">₹${p.price.toFixed(2)}</p>
89.                 <button onclick="addToCart(${p.id})">Add to Cart</button>
90.             </div>
91.         `).join('');
92.     }
93.
94.     // 3. Add to Cart Logic
95.     function addToCart(productId) {
96.         const item = cart.find(i => i.id === productId);
97.         if (item) {
98.             item.qty += 1;
99.         } else {
100.             const product = products.find(p => p.id === productId);
101.             cart.push({ ...product, qty: 1 });
102.         }
103.         renderCart();
104.     }
105.
106.     // 4. Remove from Cart Logic
107.     function removeFromCart(productId) {
108.         cart = cart.filter(i => i.id !== productId);
109.         renderCart();
110.     }
111.
112.     // 5. Render Cart Items & Total (Right Column)
113.     function renderCart() {
114.         const table = document.getElementById("cartTable");
115.         const totalEl = document.getElementById("grandTotal");
116.
117.         if (cart.length === 0) {
118.             table.innerHTML = `<tr><td colspan="4" style="text-align:center; color:#888;">Cart is empty</td></tr>`;
119.             totalEl.textContent = "₹0.00";
120.             return;
121.         }
122.
123.         table.innerHTML = cart.map(i => `
124.             <tr>
```



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```
125.         <td>${i.name}</td>
126.         <td>${i.qty}</td>
127.         <td>₹${(i.price * i.qty).toFixed(2)}</td>
128.         <td><button class="remove-btn"
onclick="removeFromCart(${i.id})">X</button></td>
129.     </tr>
130.     `).join('');
131.
132.     const total = cart.reduce((sum, item) => sum +
(item.price * item.qty), 0);
133.     totalEl.textContent = `₹${total.toFixed(2)}`;
134. }
135.
136. function checkout() {
137.     if (cart.length === 0) return alert("Your cart is
empty!");
138.     alert("Order placed successfully!");
139.     cart = [];
140.     renderCart();
141. }
142.
143. // Initial load
144. renderProducts();
145. renderCart();
146. </script>
147.
148. </body>
149. </html>
```

3.Output:

127.0.0.1:5500/Exp%202/Case_Study2_Aaryan_Kawale_1162.html

Available Items

Wireless Mouse
₹1428.00
Add to Cart

Keyboard
₹2856.00
Add to Cart

Headphones
₹4284.00
Add to Cart

USB Cable
₹761.60
Add to Cart

Shopping Cart

Item	Qty	Price	
USB Cable	1	₹761.60	X
Headphones	1	₹4284.00	X
Keyboard	1	₹2856.00	X
Wireless Mouse	1	₹1428.00	X

Total: ₹9329.60

Checkout



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Experiment No. 3

- 1. Experiment Title:** Implement control structures and form validations; create a grading system based on user-entered marks.
- 2. Software/Tools Required:** VS Code, Html/JS Compiler...
- 3. Experiment Program Code:**

```
4. <!DOCTYPE html>
5. <html lang="en">
6. <head>
7.   <meta charset="UTF-8">
8.   <meta name="viewport" content="width=device-width, initial-
   scale=1.0">
9.   <title>5-Subject Grading System</title>
10. <style>
11.   * { box-sizing: border-box; font-family: system-ui, -apple-system,
   sans-serif; }
12.   body { background-color: #f4f6f9; display: flex; justify-content:
   center; align-items: center; min-height: 100vh; margin: 0; padding:
   1.5rem; }
13.   .card { background: #ffffff; padding: 2rem; border-radius: 12px;
   box-shadow: 0 4px 16px rgba(0,0,0,0.08); width: 100%; max-width:
   480px; }
14.   h2 { margin-top: 0; color: #1e293b; text-align: center; }
15.   .grid { display: grid; grid-template-columns: 1fr 1fr; gap:
   0.75rem 1rem; }
16.   .form-group { margin-bottom: 0.75rem; }
17.   .full-width { grid-column: span 2; }
18.   label { display: block; margin-bottom: 0.35rem; font-weight: 600;
   color: #334155; font-size: 0.9rem; }
19.   input[type="number"] { width: 100%; padding: 0.65rem; border: 1px
   solid #cbd5e1; border-radius: 6px; font-size: 0.95rem; }
20.   input[type="number"]:focus { outline: none; border-color: #2563eb;
   box-shadow: 0 0 0 3px rgba(37,99,235,0.15); }
21.   button { width: 100%; padding: 0.75rem; background-color: #2563eb;
   color: white; border: none; border-radius: 6px; font-size: 1rem; font-
   weight: 600; cursor: pointer; margin-top: 0.5rem; transition:
   background 0.2s; }
22.   button:hover { background-color: #1d4ed8; }
23.   .error-banner { background-color: #fef2f2; border-left: 4px solid
   #ef4444; color: #991b1b; padding: 0.75rem; border-radius: 4px; font-
   size: 0.875rem; margin-bottom: 1rem; display: none; }
24.   .result-box { margin-top: 1.25rem; padding: 1.25rem; border-
   radius: 8px; display: none; }
```



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```
25. .result-header { font-size: 1.1rem; font-weight: 700; margin-
    bottom: 0.5rem; text-align: center; }
26. .result-details { font-size: 0.95rem; line-height: 1.5; }
27. /* Dynamic Theme Colors */
28. .grade-A { background-color: #dcfce7; color: #166534; border: 1px
    solid #bbf7d0; }
29. .grade-B { background-color: #e0f2fe; color: #075985; border: 1px
    solid #bae6fd; }
30. .grade-C { background-color: #fef9c3; color: #854d0e; border: 1px
    solid #fef08a; }
31. .grade-D { background-color: #ffedd5; color: #9a3412; border: 1px
    solid #fed7aa; }
32. .grade-F { background-color: #fee2e2; color: #991b1b; border: 1px
    solid #fca5a5; }
33. </style>
34.</head>
35.<body>
36.
37. <div class="card">
38.   <h2>Student Assessment</h2>
39.
40.   <!-- Unified Error Banner -->
41.   <div id="errorBanner" class="error-banner"></div>
42.
43.   <form id="gradeForm" novalidate>
44.     <div class="grid">
45.       <div class="form-group">
46.         <label for="sub1">Mathematics</label>
47.         <input type="number" id="sub1" class="subject-input"
            placeholder="0 - 100">
48.       </div>
49.       <div class="form-group">
50.         <label for="sub2">Science</label>
51.         <input type="number" id="sub2" class="subject-input"
            placeholder="0 - 100">
52.       </div>
53.       <div class="form-group">
54.         <label for="sub3">English</label>
55.         <input type="number" id="sub3" class="subject-input"
            placeholder="0 - 100">
56.       </div>
57.       <div class="form-group">
58.         <label for="sub4">History</label>
59.         <input type="number" id="sub4" class="subject-input"
            placeholder="0 - 100">
60.       </div>
61.       <div class="form-group full-width">
62.         <label for="sub5">Computer Science</label>
```



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```
63.     <input type="number" id="sub5" class="subject-input"
placeholder="0 - 100">
64.     </div>
65.     </div>
66.     <button type="submit">Calculate Final Results</button>
67. </form>
68.
69. <div id="resultBox" class="result-box">
70.     <div id="resultHeader" class="result-header"></div>
71.     <div id="resultDetails" class="result-details"></div>
72. </div>
73. </div>
74.
75. <script>
76.     document.getElementById('gradeForm').addEventListener('submit',
function(e) {
77.         e.preventDefault();
78.
79.         const inputs = document.querySelectorAll('.subject-input');
80.         const errorBanner = document.getElementById('errorBanner');
81.         const resultBox = document.getElementById('resultBox');
82.         const resultHeader = document.getElementById('resultHeader');
83.         const resultDetails = document.getElementById('resultDetails');
84.
85.         // Reset UI state
86.         errorBanner.style.display = 'none';
87.         resultBox.style.display = 'none';
88.         resultBox.className = 'result-box';
89.
90.         const scores = [];
91.         let hasFailedSubject = false;
92.
93.         // --- 1. CONTROL STRUCTURE: Loop with Validation Checks ---
94.         for (let i = 0; i < inputs.length; i++) {
95.             const value = inputs[i].value.trim();
96.             const label = inputs[i].previousElementSibling.textContent;
97.
98.             // Check empty input
99.             if (value === '') {
100.                 showError(`Please fill in marks for
<strong>${label}</strong>.`);
101.                 inputs[i].focus();
102.                 return;
103.             }
104.
105.             const score = Number(value);
106.
107.             // Check invalid numbers or out-of-bounds inputs
```



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```
108.         if (isNaN(score) || score < 0 || score > 100) {
109.             showError(`Invalid mark for
    <strong>${label}</strong>. Enter a value between 0 and 100.`);
110.             inputs[i].focus();
111.             return;
112.         }
113.
114.         // Check for subject failure condition (e.g., passing
    mark is 40)
115.         if (score < 40) {
116.             hasFailedSubject = true;
117.         }
118.
119.         scores.push(score);
120.     }
121.
122.     // --- 2. CALCULATIONS ---
123.     const totalMarks = scores.reduce((sum, current) => sum +
    current, 0);
124.     const percentage = (totalMarks / (scores.length * 100)) *
    100;
125.
126.     // --- 3. CONTROL STRUCTURE: Grading Logic ---
127.     let grade = '';
128.     let status = '';
129.     let cssClass = '';
130.
131.     if (hasFailedSubject) {
132.         // Essential academic condition: Failing any individual
    subject overrides average grade
133.         status = 'FAILED (Failed 1 or more subjects)';
134.         grade = 'F';
135.         cssClass = 'grade-F';
136.     } else if (percentage >= 90) {
137.         status = 'PASSED';
138.         grade = 'A+ (Outstanding)';
139.         cssClass = 'grade-A';
140.     } else if (percentage >= 80) {
141.         status = 'PASSED';
142.         grade = 'A (Excellent)';
143.         cssClass = 'grade-A';
144.     } else if (percentage >= 70) {
145.         status = 'PASSED';
146.         grade = 'B (Good)';
147.         cssClass = 'grade-B';
148.     } else if (percentage >= 60) {
149.         status = 'PASSED';
150.         grade = 'C (Satisfactory)';
```



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```
151.         cssClass = 'grade-C';
152.     } else if (percentage >= 50) {
153.         status = 'PASSED';
154.         grade = 'D (Pass)';
155.         cssClass = 'grade-D';
156.     } else {
157.         status = 'FAILED';
158.         grade = 'F (Needs Improvement)';
159.         cssClass = 'grade-F';
160.     }
161.
162.     // --- 4. RENDER OUTPUT ---
163.     resultHeader.textContent = `Status: ${status}`;
164.     resultDetails.innerHTML = `
165.         <strong>Total Marks:</strong> ${totalMarks} / 500<br>
166.         <strong>Overall Percentage:</strong>
167.         ${percentage.toFixed(2)}%<br>
168.         <strong>Final Grade:</strong> ${grade}<br>
169.         <hr style="margin: 0.75rem 0; border: none; border-top:
170.         1px dashed rgba(0,0,0,0.2);">
171.         <strong>Student Name:</strong> Aaryan K. Kawale<br>
172.         <strong>PRN:</strong> 24070521162
173.         `;
174.     resultBox.classList.add(cssClass);
175.     resultBox.style.display = 'block';
176.
177.     function showError(htmlMessage) {
178.         const errorBanner =
179.         document.getElementById('errorBanner');
180.         errorBanner.innerHTML = htmlMessage;
181.         errorBanner.style.display = 'block';
182.     }
183. </script>
184. </body>
185. </html>
```



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4.Output:

Student Grading System

5-Subject Grading System

Ask Gemini

127.0.0.1:5500/Exp%203/exp3.html

School

All Bookmarks

Student Assessment

Mathematics	Science
85	84
English	History
85	74
Computer Science	
98	

Calculate Final Results

Status: PASSED

Total Marks: 426 / 500

Overall Percentage: 85.20%

Final Grade: A (Excellent)

Student Name: Aaryan K. Kawale

PRN: 24070521162



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Case Study 3

1. Case Study Title: Desing a webpage that match password with the user password, if matched it displays result, if not result not displayed.

2. Case Study Program Code:

```
3. <!DOCTYPE html>
4. <html lang="en">
5. <head>
6.   <meta charset="UTF-8">
7.   <meta name="viewport" content="width=device-width, initial-
   scale=1.0">
8.   <title>Result Portal</title>
9.   <style>
10.    * {
11.      box-sizing: border-box;
12.      margin: 0;
13.      padding: 0;
14.      font-family: 'Segoe UI', sans-serif;
15.    }
16.
17.    body {
18.      display: flex;
19.      min-height: 100vh;
20.      background: #f8fafc;
21.    }
22.
23.    /* Split Screen Layout */
24.    .section-left {
25.      flex: 1;
26.      padding: 40px 20px;
27.      background: #ffffff;
28.      border-right: 2px solid #e2e8f0;
29.      display: flex;
30.      justify-content: center;
31.      align-items: center;
32.    }
33.
34.    .section-right {
35.      flex: 1;
36.      padding: 40px 20px;
37.      display: flex;
38.      justify-content: center;
39.      align-items: center;
```



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```
40.     }
41.
42.     /* Form Design */
43.     .box {
44.         width: 100%;
45.         max-width: 320px;
46.     }
47.
48.     h2 {
49.         margin-bottom: 20px;
50.         color: #0f172a;
51.     }
52.
53.     label {
54.         display: block;
55.         font-weight: 600;
56.         margin-bottom: 8px;
57.         color: #334155;
58.     }
59.
60.     input {
61.         width: 100%;
62.         padding: 10px 12px;
63.         border: 1px solid #cbd5e1;
64.         border-radius: 6px;
65.         font-size: 16px;
66.         margin-bottom: 15px;
67.         outline: none;
68.     }
69.
70.     input:focus {
71.         border-color: #2563eb;
72.     }
73.
74.     button {
75.         width: 100%;
76.         padding: 10px;
77.         background: #2563eb;
78.         color: white;
79.         border: none;
80.         border-radius: 6px;
81.         font-size: 16px;
82.         font-weight: 600;
83.         cursor: pointer;
84.     }
85.
86.     button:hover {
87.         background: #1d4ed8;
```



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```
88.     }
89.
90.     .error {
91.         color: #dc2626;
92.         background: #fef2f2;
93.         border: 1px solid #fecaca;
94.         padding: 8px;
95.         border-radius: 6px;
96.         margin-top: 12px;
97.         display: none;
98.         text-align: center;
99.         font-size: 14px;
100.    }
101.
102.    /* Result Card Design */
103.    .placeholder {
104.        color: #64748b;
105.        font-size: 16px;
106.    }
107.
108.    .result-card {
109.        display: none;
110.        background: white;
111.        border: 2px solid #10b981;
112.        border-radius: 12px;
113.        padding: 30px;
114.        text-align: center;
115.        box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.1);
116.        width: 260px;
117.    }
118.
119.    .status {
120.        background: #d1fae5;
121.        color: #065f46;
122.        padding: 4px 12px;
123.        border-radius: 12px;
124.        font-weight: bold;
125.        font-size: 14px;
126.        display: inline-block;
127.        margin-bottom: 15px;
128.    }
129.
130.    .gpa {
131.        font-size: 48px;
132.        font-weight: 800;
133.        color: #2563eb;
134.    }
135.
```



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```
136.         .label {
137.             color: #64748b;
138.             font-size: 14px;
139.             font-weight: 600;
140.         }
141.     </style>
142. </head>
143. <body>
144.
145.     <!-- LEFT COLUMN: PASSWORD INPUT -->
146.     <div class="section-left">
147.         <div class="box">
148.             <h2>Student Login</h2>
149.             <form id="loginForm">
150.                 <label for="passInput">Password</label>
151.                 <!-- type="text" की वजह से पासवर्ड टाइप करते वक्त दिखता
रहेगा -->
152.                 <input type="text" id="passInput" placeholder="Enter
password..." required autocomplete="off">
153.                 <button type="submit">Submit</button>
154.             </form>
155.             <div id="errorMsg" class="error">Incorrect
Password!</div>
156.         </div>
157.     </div>
158.
159.     <!-- RIGHT COLUMN: RESULT DISPLAY -->
160.     <div class="section-right">
161.         <div id="placeholder" class="placeholder">🔒 Enter correct
password to view result</div>
162.
163.         <div id="resultCard" class="result-card">
164.             <span class="status">Pass</span>
165.             <div class="gpa">9</div>
166.             <div class="label">GPA</div>
167.         </div>
168.     </div>
169.
170.     <!-- JAVASCRIPT -->
171.     <script>
172.         const TARGET_PASS = "Aaryan1162";
173.
174.         document.getElementById("loginForm").addEventListener("submit", function(e) {
175.             e.preventDefault();
176.             const inputVal =
document.getElementById("passInput").value.trim();
177.
```



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```
178.         const isCorrect = (inputVal === TARGET_PASS);
179.
180.         document.getElementById("errorMsg").style.display =
           isCorrect ? "none" : "block";
181.         document.getElementById("resultCard").style.display =
           isCorrect ? "block" : "none";
182.         document.getElementById("placeholder").style.display =
           isCorrect ? "none" : "block";
183.     });
184. </script>
185.
186. </body>
187. </html>
```

3.Output:

Student Login

Password

Aaryan1162

Submit

Pass

9

GPA

Student Login

Password

Aaryan24070521162

Submit

Incorrect Password!

Enter correct password to view result



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Experiment No. 4

- 1. Experiment Title:** Use Function types, scope, and closures; apply try and catch; build a palindrome checker.
- 2. Software/Tools Required:** VS Code, Html/JS Compiler...
- 3. Experiment Program Code:**

```
4. <!DOCTYPE html>
5. <html lang="en">
6. <head>
7.   <meta charset="UTF-8">
8.   <meta name="viewport" content="width=device-width, initial-
  scale=1.0">
9.   <title>Palindrome Checker</title>
10.  <style>
11.    :root {
12.      --bg-color: #f4f6f8;
13.      --card-bg: #ffffff;
14.      --primary: #4f46e5;
15.      --primary-hover: #4338ca;
16.      --text-main: #1f2937;
17.      --text-muted: #6b7280;
18.      --success-bg: #d1fae5;
19.      --success-text: #065f46;
20.      --error-bg: #fee2e2;
21.      --error-text: #991b1b;
22.    }
23.
24.    * {
25.      box-sizing: border-box;
26.      margin: 0;
27.      padding: 0;
28.      font-family: -apple-system, BlinkMacSystemFont, "Segoe UI",
  Roboto, Helvetica, Arial, sans-serif;
29.    }
30.
31.    body {
32.      background-color: var(--bg-color);
33.      color: var(--text-main);
34.      display: flex;
35.      justify-content: center;
36.      align-items: center;
37.      min-height: 100vh;
```



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```
38.     padding: 20px;
39.   }
40.
41.   .container {
42.     background-color: var(--card-bg);
43.     padding: 32px;
44.     border-radius: 12px;
45.     box-shadow: 0 10px 15px -3px rgba(0, 0, 0, 0.1), 0 4px 6px -2px
rgba(0, 0, 0, 0.05);
46.     width: 100%;
47.     max-width: 480px;
48.   }
49.
50.   h1 {
51.     font-size: 1.5rem;
52.     margin-bottom: 8px;
53.     color: var(--text-main);
54.     text-align: center;
55.   }
56.
57.   p.subtitle {
58.     font-size: 0.9rem;
59.     color: var(--text-muted);
60.     text-align: center;
61.     margin-bottom: 24px;
62.   }
63.
64.   .input-group {
65.     display: flex;
66.     flex-direction: column;
67.     gap: 12px;
68.   }
69.
70.   input[type="text"] {
71.     width: 100%;
72.     padding: 12px 16px;
73.     font-size: 1rem;
74.     border: 2px solid #e5e7eb;
75.     border-radius: 8px;
76.     outline: none;
77.     transition: border-color 0.2s;
78.   }
79.
80.   input[type="text"]:focus {
81.     border-color: var(--primary);
82.   }
83.
84.   button {
```



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```
85.     width: 100%;
86.     padding: 12px;
87.     font-size: 1rem;
88.     font-weight: 600;
89.     color: #ffffff;
90.     background-color: var(--primary);
91.     border: none;
92.     border-radius: 8px;
93.     cursor: pointer;
94.     transition: background-color 0.2s;
95. }
96.
97. button:hover {
98.     background-color: var(--primary-hover);
99. }
100.
101.     #result {
102.         margin-top: 20px;
103.         padding: 12px 16px;
104.         border-radius: 8px;
105.         font-size: 0.95rem;
106.         text-align: center;
107.         display: none;
108.         word-break: break-word;
109.     }
110.
111.     #result.success {
112.         display: block;
113.         background-color: var(--success-bg);
114.         color: var(--success-text);
115.     }
116.
117.     #result.error {
118.         display: block;
119.         background-color: var(--error-bg);
120.         color: var(--error-text);
121.     }
122. </style>
123. </head>
124. <body>
125.
126.     <div class="container">
127.         <h1>Palindrome Checker</h1>
128.         <p class="subtitle">Enter a word, number, or phrase to
129. test</p>
130.         <div class="input-group">
```



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```
131.         <input type="text" id="userInput" placeholder="e.g.,
           racecar, 12321, or A man, a plan..." />
132.         <button onclick="checkPalindrome()">Check</button>
133.     </div>
134.
135.     <div id="result"></div>
136. </div>
137.
138. <script>
139.     function checkPalindrome() {
140.         const inputElement =
           document.getElementById("userInput");
141.         const resultElement = document.getElementById("result");
142.         const rawText = inputElement.value;
143.
144.         if (!rawText.trim()) {
145.             resultElement.className = "error";
146.             resultElement.textContent = "Please enter something to
           check!";
147.             return;
148.         }
149.
150.         // Clean the string: remove non-alphanumeric characters
           and convert to lowercase
151.         const cleaned = rawText.toLowerCase().replace(/[^\a-z0-
           9]/g, "");
152.
153.         // Reverse the cleaned string
154.         const reversed = cleaned.split("").reverse().join("");
155.
156.         // Compare cleaned string with its reversed counterpart
157.         if (cleaned === reversed) {
158.             resultElement.className = "success";
159.             resultElement.innerHTML =
           `<strong>"${rawText}"</strong> is a palindrome!`;
160.         } else {
161.             resultElement.className = "error";
162.             resultElement.innerHTML =
           `<strong>"${rawText}"</strong> is NOT a palindrome.`;
163.         }
164.     }
165.
166.     // Allow pressing "Enter" key to trigger the check
167.     document.getElementById("userInput").addEventListener("keyp
           ress", function(event) {
168.         if (event.key === "Enter") {
169.             checkPalindrome();
170.         }
171.     });
172. }
```



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```
171.     });  
172.     </script>  
173.  
174.     </body>  
175.     </html>
```

4.Output;

Palindrome Checker
Enter a word, number, or phrase to test

Check

"1162-2611" is a palindrome!

Palindrome Checker
Enter a word, number, or phrase to test

Check

"Aaryan" is NOT a palindrome.



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Case Study 4

1. Case Study Title: Desing a webpage to check whether the password is Palindrome or not if Palindrome display msg Weak Pass if not display Strong Pass.

2. Case Study Program Code:

```
3. <!DOCTYPE html>
4. <html lang="en">
5. <head>
6.   <meta charset="UTF-8">
7.   <meta name="viewport" content="width=device-width, initial-
  scale=1.0">
8.   <title>Password Palindrome & Strength Checker</title>
9.   <style>
10.    :root {
11.      --bg-color: #0f172a;
12.      --card-bg: #1e293b;
13.      --primary: #6366f1;
14.      --primary-hover: #4f46e5;
15.      --text-main: #f8fafc;
16.      --text-muted: #94a3b8;
17.      --warning-bg: #451a03;
18.      --warning-text: #fde047;
19.      --warning-border: #854d0e;
20.      --success-bg: #064e3b;
21.      --success-text: #6ee7b7;
22.      --success-border: #047857;
23.      --error-bg: #450a0a;
24.      --error-text: #fca5a5;
25.      --error-border: #991b1b;
26.    }
27.
28.    * {
29.      box-sizing: border-box;
30.      margin: 0;
31.      padding: 0;
32.      font-family: -apple-system, BlinkMacSystemFont, "Segoe UI",
  Roboto, Helvetica, Arial, sans-serif;
33.    }
34.
35.    body {
36.      background-color: var(--bg-color);
37.      color: var(--text-main);
38.      display: flex;
39.      justify-content: center;
```



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```
40.     align-items: center;
41.     min-height: 100vh;
42.     padding: 20px;
43. }
44.
45. .container {
46.     background-color: var(--card-bg);
47.     padding: 32px;
48.     border-radius: 12px;
49.     box-shadow: 0 10px 25px -5px rgba(0, 0, 0, 0.5);
50.     width: 100%;
51.     max-width: 480px;
52.     border: 1px solid #334155;
53. }
54.
55. h1 {
56.     font-size: 1.5rem;
57.     margin-bottom: 8px;
58.     color: var(--text-main);
59.     text-align: center;
60. }
61.
62. p.subtitle {
63.     font-size: 0.9rem;
64.     color: var(--text-muted);
65.     text-align: center;
66.     margin-bottom: 24px;
67. }
68.
69. .input-group {
70.     display: flex;
71.     flex-direction: column;
72.     gap: 12px;
73. }
74.
75. .password-wrapper {
76.     position: relative;
77.     width: 100%;
78.     display: flex;
79.     align-items: center;
80. }
81.
82. input[type="text"] {
83.     width: 100%;
84.     padding: 12px 60px 12px 16px;
85.     font-size: 1.1rem;
86.     letter-spacing: 1px;
87.     background-color: #0f172a;
```



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```
88.     color: var(--text-main);
89.     border: 2px solid #334155;
90.     border-radius: 8px;
91.     outline: none;
92.     transition: border-color 0.2s;
93. }
94.
95. input:focus {
96.     border-color: var(--primary);
97. }
98.
99. .toggle-btn {
100.     position: absolute;
101.     right: 12px;
102.     background: none;
103.     border: none;
104.     color: var(--text-muted);
105.     cursor: pointer;
106.     font-size: 0.85rem;
107.     padding: 4px 8px;
108.     border-radius: 4px;
109.     width: auto;
110. }
111.
112. .toggle-btn:hover {
113.     color: var(--text-main);
114.     background-color: transparent;
115. }
116.
117. button.btn-submit {
118.     width: 100%;
119.     padding: 12px;
120.     font-size: 1rem;
121.     font-weight: 600;
122.     color: #ffffff;
123.     background-color: var(--primary);
124.     border: none;
125.     border-radius: 8px;
126.     cursor: pointer;
127.     transition: background-color 0.2s;
128. }
129.
130. button.btn-submit:hover {
131.     background-color: var(--primary-hover);
132. }
133.
134. #result {
135.     margin-top: 20px;
```



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```
136.         padding: 14px 16px;
137.         border-radius: 8px;
138.         font-size: 0.95rem;
139.         text-align: center;
140.         display: none;
141.         word-break: break-word;
142.         border: 1px solid transparent;
143.     }
144.
145.     #result.warning {
146.         display: block;
147.         background-color: var(--warning-bg);
148.         color: var(--warning-text);
149.         border-color: var(--warning-border);
150.     }
151.
152.     #result.success {
153.         display: block;
154.         background-color: var(--success-bg);
155.         color: var(--success-text);
156.         border-color: var(--success-border);
157.     }
158.
159.     #result.error {
160.         display: block;
161.         background-color: var(--error-bg);
162.         color: var(--error-text);
163.         border-color: var(--error-border);
164.     }
165. </style>
166. </head>
167. <body>
168.
169.     <div class="container">
170.         <h1>Password Security Checker</h1>
171.         <p class="subtitle">Check if your password is a vulnerable
172.         palindrome</p>
173.
174.         <div class="input-group">
175.             <div class="password-wrapper">
176.                 <input type="text" id="passwordInput"
177.                 placeholder="Enter password..." />
178.                 <button type="button" class="toggle-btn" id="toggleBtn"
179.                 onclick="toggleVisibility()">Hide</button>
180.             </div>
181.             <button class="btn-submit"
182.             onclick="checkPassword()">Check Password</button>
183.         </div>
184.     </div>
```



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```
180.
181.     <div id="result"></div>
182. </div>
183.
184.     <script>
185.         let actualPassword = "";
186.         let isHidden = false;
187.
188.         const inputElement =
            document.getElementById("passwordInput");
189.
190.         // Intercept input changes to support custom asterisk
            masking
191.         inputElement.addEventListener("input", function(e) {
192.             if (isHidden) {
193.                 // If characters were added or deleted
194.                 const diff = e.target.value.length -
                    actualPassword.length;
195.
196.                 if (diff > 0) {
197.                     // Characters added
198.                     const addedChars =
199.                         e.target.value.slice(actualPassword.length);
200.                     actualPassword += addedChars;
201.                 } else if (diff < 0) {
202.                     // Characters removed (Backspace)
203.                     actualPassword = actualPassword.slice(0,
204.                         e.target.value.length);
205.                 }
206.
207.                 // Render asterisks in the input display
208.                 inputElement.value = "*".repeat(actualPassword.length);
209.             } else {
210.                 actualPassword = e.target.value;
211.             }
212.         });
213.
214.         function toggleVisibility() {
215.             const btn = document.getElementById("toggleBtn");
216.
217.             if (!isHidden) {
218.                 // Switch to Hidden mode -> replace characters with *
219.                 isHidden = true;
220.                 inputElement.value = "*".repeat(actualPassword.length);
221.                 btn.textContent = "Show";
222.             } else {
223.                 // Switch to Visible mode -> show raw password
224.                 isHidden = false;
```



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```
223.         inputElement.value = actualPassword;
224.         btn.textContent = "Hide";
225.     }
226. }
227.
228. function checkPassword() {
229.     const resultElement = document.getElementById("result");
230.     const password = actualPassword;
231.
232.     if (!password) {
233.         resultElement.className = "error";
234.         resultElement.textContent = "Please enter a password
first!";
235.         return;
236.     }
237.
238.     // Reverse exact password
239.     const reversedPassword =
password.split("").reverse().join("");
240.
241.     // Check if exact palindrome
242.     if (password === reversedPassword) {
243.         resultElement.className = "warning";
244.         resultElement.innerHTML = "⚠️ <strong>Weak
password!</strong> It is a palindrome. Choose a strong password.";
245.     } else {
246.         resultElement.className = "success";
247.         resultElement.innerHTML = "🔒 <strong>It is a strong
non-palindrome password.</strong>";
248.     }
249. }
250.
251. // Trigger check on Enter key press
252. inputElement.addEventListener("keypress", function(event) {
253.     if (event.key === "Enter") {
254.         checkPassword();
255.     }
256. });
257. </script>
258.
259. </body>
260. </html>
```



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3.Output:

The screenshot shows a web browser window with the title "Password Palindrome & Strength". The address bar shows the URL "127.0.0.1:5500/Exp4/exp4_Aaryan_Kawale_1162CS.html". The page content features a "Password Security Checker" form. The input field contains the password "Aaryan". Below the input field is a "Check Password" button. The output message, displayed in a green box, states: "It is a strong non-palindrome password."

The screenshot shows the same web browser window. The input field now contains the password "1162-2611". The "Check Password" button is still present. The output message, displayed in a brown box, states: "Weak password! It is a palindrome. Choose a strong password."



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Case Study

1. Case Study Title: Desing a webpage to check whether the password is Palindrome or not if Palindrome display msg Weak Pass if not display Strong Pass.

2. Case Study Program Code:

```
3. <!DOCTYPE html>
4. <html lang="en">
5. <head>
6.   <meta charset="UTF-8">
7.   <meta name="viewport" content="width=device-width, initial-
  scale=1.0">
8.   <title>Password Palindrome & Strength Checker</title>
9.   <style>
10.    :root {
11.      --bg-color: #0f172a;
12.      --card-bg: #1e293b;
13.      --primary: #6366f1;
14.      --primary-hover: #4f46e5;
15.      --text-main: #f8fafc;
16.      --text-muted: #94a3b8;
17.      --warning-bg: #451a03;
18.      --warning-text: #fde047;
19.      --warning-border: #854d0e;
20.      --success-bg: #064e3b;
21.      --success-text: #6ee7b7;
22.      --success-border: #047857;
23.      --error-bg: #450a0a;
24.      --error-text: #fca5a5;
25.      --error-border: #991b1b;
26.    }
27.
28.    * {
29.      box-sizing: border-box;
30.      margin: 0;
31.      padding: 0;
32.      font-family: -apple-system, BlinkMacSystemFont, "Segoe UI",
  Roboto, Helvetica, Arial, sans-serif;
33.    }
34.
35.    body {
36.      background-color: var(--bg-color);
37.      color: var(--text-main);
38.      display: flex;
39.      justify-content: center;
```



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```
40.     align-items: center;
41.     min-height: 100vh;
42.     padding: 20px;
43. }
44.
45. .container {
46.     background-color: var(--card-bg);
47.     padding: 32px;
48.     border-radius: 12px;
49.     box-shadow: 0 10px 25px -5px rgba(0, 0, 0, 0.5);
50.     width: 100%;
51.     max-width: 480px;
52.     border: 1px solid #334155;
53. }
54.
55. h1 {
56.     font-size: 1.5rem;
57.     margin-bottom: 8px;
58.     color: var(--text-main);
59.     text-align: center;
60. }
61.
62. p.subtitle {
63.     font-size: 0.9rem;
64.     color: var(--text-muted);
65.     text-align: center;
66.     margin-bottom: 24px;
67. }
68.
69. .input-group {
70.     display: flex;
71.     flex-direction: column;
72.     gap: 12px;
73. }
74.
75. .password-wrapper {
76.     position: relative;
77.     width: 100%;
78.     display: flex;
79.     align-items: center;
80. }
81.
82. input[type="text"] {
83.     width: 100%;
84.     padding: 12px 60px 12px 16px;
85.     font-size: 1.1rem;
86.     letter-spacing: 1px;
87.     background-color: #0f172a;
```



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```
88.     color: var(--text-main);
89.     border: 2px solid #334155;
90.     border-radius: 8px;
91.     outline: none;
92.     transition: border-color 0.2s;
93. }
94.
95. input:focus {
96.     border-color: var(--primary);
97. }
98.
99. .toggle-btn {
100.     position: absolute;
101.     right: 12px;
102.     background: none;
103.     border: none;
104.     color: var(--text-muted);
105.     cursor: pointer;
106.     font-size: 0.85rem;
107.     padding: 4px 8px;
108.     border-radius: 4px;
109.     width: auto;
110. }
111.
112. .toggle-btn:hover {
113.     color: var(--text-main);
114.     background-color: transparent;
115. }
116.
117. button.btn-submit {
118.     width: 100%;
119.     padding: 12px;
120.     font-size: 1rem;
121.     font-weight: 600;
122.     color: #ffffff;
123.     background-color: var(--primary);
124.     border: none;
125.     border-radius: 8px;
126.     cursor: pointer;
127.     transition: background-color 0.2s;
128. }
129.
130. button.btn-submit:hover {
131.     background-color: var(--primary-hover);
132. }
133.
134. #result {
135.     margin-top: 20px;
```



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```
136.         padding: 14px 16px;
137.         border-radius: 8px;
138.         font-size: 0.95rem;
139.         text-align: center;
140.         display: none;
141.         word-break: break-word;
142.         border: 1px solid transparent;
143.     }
144.
145.     #result.warning {
146.         display: block;
147.         background-color: var(--warning-bg);
148.         color: var(--warning-text);
149.         border-color: var(--warning-border);
150.     }
151.
152.     #result.success {
153.         display: block;
154.         background-color: var(--success-bg);
155.         color: var(--success-text);
156.         border-color: var(--success-border);
157.     }
158.
159.     #result.error {
160.         display: block;
161.         background-color: var(--error-bg);
162.         color: var(--error-text);
163.         border-color: var(--error-border);
164.     }
165. </style>
166. </head>
167. <body>
168.
169.     <div class="container">
170.         <h1>Password Security Checker</h1>
171.         <p class="subtitle">Check if your password is a vulnerable
palindrome</p>
172.
173.         <div class="input-group">
174.             <div class="password-wrapper">
175.                 <input type="text" id="passwordInput"
placeholder="Enter password..." />
176.                 <button type="button" class="toggle-btn" id="toggleBtn"
onclick="toggleVisibility()">Hide</button>
177.             </div>
178.             <button class="btn-submit"
onclick="checkPassword()">Check Password</button>
179.         </div>
```



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```
180.
181.     <div id="result"></div>
182. </div>
183.
184.     <script>
185.         let actualPassword = "";
186.         let isHidden = false;
187.
188.         const inputElement =
189.             document.getElementById("passwordInput");
190.         // Intercept input changes to support custom asterisk
191.         masking
192.         inputElement.addEventListener("input", function(e) {
193.             if (isHidden) {
194.                 // If characters were added or deleted
195.                 const diff = e.target.value.length -
196.                     actualPassword.length;
197.                 if (diff > 0) {
198.                     // Characters added
199.                     const addedChars =
200.                         e.target.value.slice(actualPassword.length);
201.                     actualPassword += addedChars;
202.                 } else if (diff < 0) {
203.                     // Characters removed (Backspace)
204.                     actualPassword = actualPassword.slice(0,
205.                         e.target.value.length);
206.                 }
207.                 // Render asterisks in the input display
208.                 inputElement.value = "*".repeat(actualPassword.length);
209.             } else {
210.                 actualPassword = e.target.value;
211.             }
212.         });
213.
214.         function toggleVisibility() {
215.             const btn = document.getElementById("toggleBtn");
216.
217.             if (!isHidden) {
218.                 // Switch to Hidden mode -> replace characters with *
219.                 isHidden = true;
220.                 inputElement.value = "*".repeat(actualPassword.length);
221.                 btn.textContent = "Show";
222.             } else {
223.                 // Switch to Visible mode -> show raw password
224.                 isHidden = false;
```



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```
223.         inputElement.value = actualPassword;
224.         btn.textContent = "Hide";
225.     }
226. }
227.
228.     function checkPassword() {
229.         const resultElement = document.getElementById("result");
230.         const password = actualPassword;
231.
232.         if (!password) {
233.             resultElement.className = "error";
234.             resultElement.textContent = "Please enter a password
first!";
235.             return;
236.         }
237.
238.         // Reverse exact password
239.         const reversedPassword =
password.split("").reverse().join("");
240.
241.         // Check if exact palindrome
242.         if (password === reversedPassword) {
243.             resultElement.className = "warning";
244.             resultElement.innerHTML = "⚠ <strong>Weak
password!</strong> It is a palindrome. Choose a strong password.";
245.         } else {
246.             resultElement.className = "success";
247.             resultElement.innerHTML = "🔒 <strong>It is a strong
non-palindrome password.</strong>";
248.         }
249.     }
250.
251.     // Trigger check on Enter key press
252.     inputElement.addEventListener("keypress", function(event) {
253.         if (event.key === "Enter") {
254.             checkPassword();
255.         }
256.     });
257. </script>
258.
259. </body>
260. </html>
```



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3.Output:



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Experiment No. 5

- Experiment Title:** Array methods and objects handling;
Create a cart total calculator with discount logic.
- Software/Tools Required:** VS Code, Html/JS Compiler...
- Experiment Program Code:**

```
4. <!DOCTYPE html>
5. <html lang="en">
6. <head>
7.   <meta charset="UTF-8">
8.   <title>MetaHat Store</title>
9.   <style>
10.    * { box-sizing: border-box; margin: 0; padding: 0; font-family:
    'Segoe UI', sans-serif; }
11.
12.    body {
13.      background-color: #f0f2f5;
14.      min-height: 100vh;
15.      display: flex;
16.      flex-direction: column;
17.      align-items: center;
18.    }
19.
20.    h1 {
21.      margin: 20px 0;
22.      color: #1a1a1a;
23.    }
24.
25.    /* Main Container: Split Left and Right Centered */
26.    .page-wrapper {
27.      display: flex;
28.      justify-content: center;
29.      align-items: flex-start;
30.      gap: 30px;
31.      width: 90%;
32.      max-width: 1000px;
33.      margin: auto 0;
34.    }
35.
36.    /* Left Box - Products Container */
37.    .box-left {
38.      flex: 1;
39.      background: #ffffff;
40.      border: 2px solid #e0e0e0;
41.      border-radius: 12px;
```



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```
42.     padding: 20px;
43.     box-shadow: 0 4px 12px rgba(0,0,0,0.05);
44. }
45.
46. /* Right Box - Shopping Cart Container */
47. .box-right {
48.     width: 320px;
49.     background: #ffffff;
50.     border: 2px solid #e0e0e0;
51.     border-radius: 12px;
52.     padding: 20px;
53.     box-shadow: 0 4px 12px rgba(0,0,0,0.05);
54. }
55.
56. .box-title {
57.     font-size: 1.2rem;
58.     font-weight: 600;
59.     margin-bottom: 15px;
60.     border-bottom: 2px solid #f0f2f5;
61.     padding-bottom: 8px;
62. }
63.
64. /* Items Grid Inside Left Box */
65. .product-grid {
66.     display: grid;
67.     grid-template-columns: repeat(auto-fit, minmax(130px, 1fr));
68.     gap: 12px;
69. }
70.
71. .product-card {
72.     border: 1px solid #e2e8f0;
73.     border-radius: 8px;
74.     padding: 12px;
75.     text-align: center;
76.     background: #fafafa;
77. }
78.
79. .product-card h4 { font-size: 0.95rem; margin-bottom: 5px; }
80. .product-card .price { color: #2563eb; font-weight: bold; margin-
    bottom: 10px; }
81.
82. button {
83.     background-color: #2563eb;
84.     color: white;
85.     border: none;
86.     padding: 6px 10px;
87.     border-radius: 4px;
88.     cursor: pointer;
```



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```
89.     font-size: 0.85rem;
90.   }
91.
92.   button:hover { background-color: #1d4ed8; }
93.
94.   /* Cart Items List */
95.   .cart-items {
96.     max-height: 200px;
97.     overflow-y: auto;
98.     margin-bottom: 15px;
99.   }
100.
101.   .cart-item {
102.     display: flex;
103.     justify-content: space-between;
104.     align-items: center;
105.     padding: 8px 0;
106.     border-bottom: 1px dashed #e2e8f0;
107.     font-size: 0.9rem;
108.   }
109.
110.   .btn-remove {
111.     background: #ef4444;
112.     padding: 2px 6px;
113.     font-size: 0.75rem;
114.   }
115.   .btn-remove:hover { background: #dc2626; }
116.
117.   /* Price Calculation Block */
118.   .price-details {
119.     border-top: 2px solid #e2e8f0;
120.     padding-top: 10px;
121.     font-size: 0.95rem;
122.   }
123.
124.   .price-row {
125.     display: flex;
126.     justify-content: space-between;
127.     margin-bottom: 8px;
128.   }
129.
130.   .discount-text { color: #16a34a; font-weight: 600; }
131.
132.   .final-row {
133.     font-size: 1.15rem;
134.     font-weight: bold;
135.     border-top: 1px solid #e2e8f0;
136.     padding-top: 8px;
```



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```
137.         margin-top: 8px;
138.     }
139. </style>
140. </head>
141. <body>
142.
143.     <h1>Insta Mart</h1>
144.
145.     <div class="page-wrapper">
146.
147.         <!-- LEFT BOX: Item Listing -->
148.         <div class="box-left">
149.             <div class="box-title">Available Items</div>
150.             <div id="product-list" class="product-grid"><div>
151.         </div>
152.
153.         <!-- RIGHT BOX: Cart & Price Calculations -->
154.         <div class="box-right">
155.             <div class="box-title">Your Cart</div>
156.
157.             <div id="cart-items" class="cart-items"></div>
158.
159.             <div class="price-details">
160.                 <div class="price-row">
161.                     <span>Cart Price:</span>
162.                     <span>₹<span id="cart-price">0</span></span></span>
163.                 </div>
164.                 <div class="price-row discount-text">
165.                     <span>Discount (<span id="discount-
percent">0</span>%):</span></span>
166.                     <span>-₹<span id="discount-price">0</span></span></span>
167.                 </div>
168.                 <div class="price-row final-row">
169.                     <span>Final Price:</span>
170.                     <span>₹<span id="final-price">0</span></span></span>
171.                 </div>
172.             </div>
173.         </div>
174.
175.     </div>
176.
177.     <script>
178.         //
=====
179.         // ✨ HIGHLIGHT: ITEM LIST ARRAY (EDIT/ADD YOUR ITEMS HERE)
180.         //
=====
```



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```
181.     const products = [
182.         { id: 1, name: "MetaCap", price: 400 },
183.         { id: 2, name: "Cyber Hat", price: 800 },
184.         { id: 4, name: "Meta Glasses", price: 3200 },
185.         { id: 6, name: "Watch", price: 2000 },
186.         { id: 7, name: "Stylus", price: 500 },
187.         { id: 8, name: "Pen drive", price: 200 },
188.         { id: 9, name: "VR Visor", price: 1200 },
189.         { id: 10, name: "Smart Crown", price: 2500 }
190.     ];
191.     //
=====
192.
193.     let cart = [];
194.
195.     // Render items inside Left Box
196.     function renderProducts() {
197.         document.getElementById('product-list').innerHTML =
products.map(p => `
198.             <div class="product-card">
199.                 <h4>${p.name}</h4>
200.                 <div class="price">₹${p.price}</div>
201.                 <button onclick="addToCart(${p.id})">Add to
Cart</button>
202.             </div>
203.         `).join('');
204.     }
205.
206.     // Add item to cart
207.     function addToCart(id) {
208.         const item = products.find(p => p.id === id);
209.         cart.push(item);
210.         updateCart();
211.     }
212.
213.     // Remove item from cart
214.     function removeFromCart(index) {
215.         cart.splice(index, 1);
216.         updateCart();
217.     }
218.
219.     // Calculate prices and update Right Box UI
220.     function updateCart() {
221.         const cartContainer = document.getElementById('cart-
items');
222.
223.         if (cart.length === 0) {
```



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```
224.         cartContainer.innerHTML = '<p style="color: #94a3b8;
text-align: center;">Cart is empty</p>';
225.     } else {
226.         cartContainer.innerHTML = cart.map((item, index) => `
227.             <div class="cart-item">
228.                 <span>${item.name} (₹${item.price})</span>
229.                 <button class="btn-remove"
onclick="removeFromCart(${index})">Remove</button>
230.             </div>
231.         `).join('');
232.     }
233.
234.     // 1. Calculate Cart Subtotal
235.     const cartPrice = cart.reduce((sum, item) => sum +
item.price, 0);
236.
237.     // 2. Discount Tier Rules
238.     let discountPercent = 0;
239.     if (cartPrice > 3000) discountPercent = 30;
240.     else if (cartPrice > 2000) discountPercent = 20;
241.     else if (cartPrice > 1000) discountPercent = 10;
242.
243.     // 3. Discount Calculations
244.     const discountPrice = (cartPrice * discountPercent) /
100;
245.     const finalPrice = cartPrice - discountPrice;
246.
247.     // Render Values
248.     document.getElementById('cart-price').textContent =
cartPrice;
249.     document.getElementById('discount-percent').textContent =
discountPercent;
250.     document.getElementById('discount-price').textContent =
discountPrice;
251.     document.getElementById('final-price').textContent =
finalPrice;
252. }
253.
254. // Initialize Page
255. renderProducts();
256. updateCart();
257. </script>
258.
259. </body>
260. </html>
```



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4.Output:

The screenshot shows a web browser window with the URL `127.0.0.1:5500/Exp5/exp5_Aaryan_Kawale_1162.html`. The page is titled "Insta Mart". On the left, under "Available Items", there are six items in a 2x3 grid: MetaCap (₹400), Cyber Hat (₹800), Meta Glasses (₹3200), Watch (₹2000), Stylus (₹500), and Pen drive (₹800). Each item has an "Add to Cart" button. On the right, under "Your Cart", there is one item: Watch (₹2000) with a "Remove" button. Below the cart item, the "Cart Price" is ₹2000, the "Discount (10%)" is -₹200, and the "Final Price" is ₹1800.

The screenshot shows the same "Insta Mart" website. In the "Available Items" section, all six items (MetaCap, Cyber Hat, Meta Glasses, Watch, Stylus, and Pen drive) are still present with their "Add to Cart" buttons. In the "Your Cart" section, there are now two items: Watch (₹2000) and Meta Glasses (₹3200), each with a "Remove" button. Below the items, the "Cart Price" is ₹5200, the "Discount (30%)" is -₹1560, and the "Final Price" is ₹3640.



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Case Study 5

1. **Case Study Title:** To design and implement an interactive, client-side web application using HTML, CSS, and JavaScript for text analysis, string manipulation, and pattern matching (regex).

2. Case Study Program Code:

```
3.
4. <!DOCTYPE html>
5. <html lang="en">
6. <head>
7. <meta charset="UTF-8">
8. <meta name="viewport" content="width=device-width,initial-scale=1">
9. <title>Text & Email Analyzer</title>
10.
11. <style>
12. *{box-sizing:border-box}
13. body{
14.     margin:0;
15.     padding:15px;
16.     font-family:Arial,sans-serif;
17.     background:linear-gradient(135deg,#667eea,#764ba2);
18. }
19. .container{
20.     max-width:1100px;
21.     margin:auto;
22. }
23. h1{
24.     color:white;
25.     text-align:center;
26.     margin:5px 0 15px;
27. }
28. .main{
29.     display:grid;
30.     grid-template-columns:1fr 1fr;
31.     gap:15px;
32. }
33. .card{
34.     background:white;
35.     padding:18px;
36.     border-radius:14px;
37.     box-shadow:0 6px 20px #0002;
38. }
39. h2{
40.     margin:0 0 12px;
41.     font-size:20px;
```



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```
42.     color:#333;
43. }
44. textarea{
45.     width:100%;
46.     height:190px;
47.     resize:none;
48.     padding:12px;
49.     border:2px solid #ddd;
50.     border-radius:9px;
51.     font-size:14px;
52.     outline:none;
53. }
54. textarea:focus{border-color:#667eea}
55. .buttons{
56.     display:flex;
57.     gap:7px;
58.     margin-top:9px;
59. }
60. button{
61.     border:0;
62.     padding:9px 13px;
63.     border-radius:7px;
64.     background:#667eea;
65.     color:white;
66.     font-weight:bold;
67.     cursor:pointer;
68. }
69. button:hover{background:#d0c74f}
70. .clear{background:#ef4444}
71. .results{
72.     display:grid;
73.     grid-template-columns:1fr 1fr;
74.     gap:8px;
75. }
76. .box{
77.     background:#f4f6ff;
78.     padding:10px;
79.     border-radius:8px;
80. }
81. .box span{
82.     display:block;
83.     color:#666;
84.     font-size:12px;
85. }
86. .box strong{
87.     font-size:16px;
88.     color:#222;
89. }
```



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```
90..full{grid-column:1/-1}
91..email{
92.    color:#70ca38;
93.    margin:3px 0;
94.    font-size:14px;
95.    word-break:break-all;
96.}
97..output{
98.    height:75px;
99.    margin-top:7px;
100.   }
101.   .search{
102.       display:flex;
103.       gap:6px;
104.       margin-top:8px;
105.   }
106.   .search input{
107.       flex:1;
108.       padding:9px;
109.       border:1px solid #ccc;
110.       border-radius:7px;
111.   }
112.   @media(max-width:700px){
113.       .main{grid-template-columns:1fr}
114.   }
115. </style>
116. </head>
117.
118. <body>
119.
120.   <div class="container">
121.
122.     <h1>📧 Text & Email Analyzer</h1>
123.
124.     <div class="main">
125.
126.       <!-- INPUT -->
127.       <div class="card">
128.         <h2>📧 Input Text</h2>
129.
130.         <textarea id="text" placeholder="Enter your paragraph
here..."></textarea>
131.
132.         <div class="buttons">
133.           <button onclick="analyze()">Analyze</button>
134.           <button onclick="swap()">Swap Case</button>
135.           <button class="clear" onclick="clearAll()">Clear</button>
136.         </div>
```



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```
137.     </div>
138.
139.     <!-- ANALYSIS -->
140.     <div class="card">
141.     <h2>📊 Analysis</h2>
142.
143.     <div class="results">
144.
145.         <div class="box">
146.         <span>Text Length</span>
147.         <strong id="length">0</strong>
148.         </div>
149.
150.         <div class="box">
151.         <span>Words</span>
152.         <strong id="words">0</strong>
153.         </div>
154.
155.         <div class="box">
156.         <span>Contains "Welcome"</span>
157.         <strong id="welcome">No</strong>
158.         </div>
159.
160.         <div class="box">
161.         <span>Email Occurrences</span>
162.         <strong id="emailOcc">0</strong>
163.         </div>
164.
165.         <div class="box full">
166.         <span>Emails Found</span>
167.         <div id="emails">None</div>
168.         </div>
169.
170.         <div class="box full">
171.         <span>Valid Email</span>
172.         <strong id="valid">No</strong>
173.         </div>
174.
175.     </div>
176.
177.     <div class="search">
178.     <input id="search" placeholder="Search a word...">
179.     <button onclick="findWord()">Search</button>
180.     </div>
181.
182.     <div class="box" style="margin-top:8px">
183.     <strong id="searchResult">Search result appears here</strong>
```



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```
184.     </div>
185.
186.     </div>
187.
188.     <!-- STRING FUNCTIONS -->
189.     <div class="card">
190.     <h2> String Functions</h2>
191.
192.     <label>UPPERCASE</label>
193.     <textarea class="output" id="upper" readonly></textarea>
194.
195.     <label>lowercase</label>
196.     <textarea class="output" id="lower" readonly></textarea>
197.
198.     </div>
199.
200.     <!-- SWAP CASE -->
201.     <div class="card">
202.     <h2> Swap Case</h2>
203.
204.     <textarea class="output" id="swapped" readonly
205.     placeholder="A → a | a → A"></textarea>
206.
207.     </div>
208.
209.     </div>
210.     </div>
211.
212.     <script>
213.
214.     const emailRegex =
215.     /[a-zA-Z0-9.!#$%&'*/+=?^_`{|}~-]+@[a-zA-Z0-9-]+(?:\.[a-zA-Z0-9-
216.     ]+)+/g;
217.
218.     // ANALYZE
219.     function analyze(){
220.
221.         const text=document.getElementById("text").value;
222.
223.         if(!text.trim()){
224.             alert("Please enter some text.");
225.             return;
226.         }
227.         document.getElementById("upper").value=text.toUpperCase();
```



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```
228.     document.getElementById("lower").value=text.toLowerCase();
229.
230.     document.getElementById("swapped").value=swapText(text);
231.
232.     document.getElementById("length").textContent=text.length;
233.
234.     document.getElementById("words").textContent=
235.         text.trim().split(/\s+/).length;
236.
237.     document.getElementById("welcome").textContent=
238.         text.toLowerCase().includes("welcome") ? "Yes" : "No";
239.
240.     const emails=text.match(emailRegex)||[];
241.
242.     document.getElementById("emailOcc").textContent=emails.length;
243.
244.     document.getElementById("valid").textContent=
245.         emails.length ? "Yes" : "No";
246.
247.     document.getElementById("emails").innerHTML=
248.         emails.length
249.         ? emails.map(e=><div class="email">✉
250.             ${e}</div>` ).join("")
251.         : "None";
252.
253.     // SWAP CASE
254.     function swapText(text){
255.         let result="";
256.
257.         for(let c of text){
258.
259.             if(c.toUpperCase()===c &&
260.                 c.toLowerCase()!==c){
261.
262.                 result+=c.toLowerCase();
263.
264.             }else if(
265.                 c.toLowerCase()===c &&
266.                 c.toUpperCase()!==c
267.             ){
268.
269.                 result+=c.toUpperCase();
270.
271.             }else{
272.
```



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```
273.         result+=c;
274.     }
275. }
276.
277.     return result;
278. }
279.
280. function swap(){
281.
282.     const text=document.getElementById("text").value;
283.
284.     document.getElementById("swapped").value=
285.         swapText(text);
286. }
287.
288. // SEARCH
289. function findWord(){
290.
291.     const text=document.getElementById("text").value;
292.     const word=document.getElementById("search").value.trim();
293.
294.     if(!word){
295.         document.getElementById("searchResult").textContent=
296.             "Enter a word.";
297.         return;
298.     }
299.
300.     const matches=
301.         text.match(new RegExp(
302.             word.replace(/[\.*+?^${}()|[\]\[\]/g,"\\$&"),
303.             "gi"
304.         )) || [];
305.
306.     document.getElementById("searchResult").textContent=
307.         `${word} found ${matches.length} time(s).`;
308. }
309.
310. // CLEAR
311. function clearAll(){
312.
313.     document.getElementById("text").value="";
314.     document.getElementById("upper").value="";
315.     document.getElementById("lower").value="";
316.     document.getElementById("swapped").value="";
317.     document.getElementById("search").value="";
318. }
```



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```
319.     document.getElementById("length").textContent="0";
320.     document.getElementById("words").textContent="0";
321.     document.getElementById("welcome").textContent="No";
322.     document.getElementById("emailOcc").textContent="0";
323.     document.getElementById("valid").textContent="No";
324.     document.getElementById("emails").textContent="None";
325.
326.     document.getElementById("searchResult").textContent=
327.         "Search result appears here";
328. }
329.
330. </script>
331.
332. </body>
333. </html>
334. Exp6.html
335. Displaying Exp6.html.
```

3.Output:

The screenshot shows a web browser displaying a web application titled "Text & Email Analyzer". The interface is divided into several sections:

- Input Text:** A text area containing the input: "My name is Aaryan Kawale 3rd year student at symbiosis Institute of tech". Below the text area are three buttons: "Analyze", "Swap Case", and "Clear".
- Analysis:** A section displaying the results of the analysis:
 - Text Length: 73
 - Words: 13
 - Contains "Welcome": No
 - Email Occurrences: 0
 - Emails Found: None
 - Valid Email: NoBelow this is a search bar with the placeholder "Search a word..." and a "Search" button. The search result area is currently empty.
- String Functions:** A section with two text areas:
 - UPPERCASE:** The input text is converted to uppercase: "MY NAME IS AARYAN KAWALE 3RD YEAR STUDENT AT SYMBIOSIS INSTITUTE OF TECH".
 - lowercase:** The input text is converted to lowercase: "my name is aaryan kawale 3rd year student at symbiosis institute of tech".
- Swap Case:** A text area showing the input text with the first letter of each word swapped: "mY NAME IS aARYAN kaWALE 3RD YEAR STUDENT AT SYMBIOSIS INSTITUTE OF TECH".



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Experiment No. 6

- 1. Experiment Title:** Use string functions and regex for email validation, data extraction, and text analysis.
- 2. Software/Tools Required:** VS Code, Html/JS Compiler...
- 3. Experiment Program Code:**

```
4. <!DOCTYPE html>
5. <html lang="en">
6. <head>
7.     <meta charset="UTF-8">
8.     <meta name="viewport" content="width=device-width, initial-
    scale=1.0">
9.     <title>Content Inspector & Email Parser</title>
10.    <style>
11.        * {
12.            box-sizing: border-box;
13.        }
14.        body {
15.            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-
    serif;
16.            background: linear-gradient(135deg, #eef2f3, #8e9eab);
17.            margin: 0;
18.            padding: 40px 20px;
19.            color: #2d3748;
20.        }
21.        .wrapper {
22.            max-width: 650px;
23.            margin: 0 auto;
24.            background: #ffffff;
25.            padding: 25px 30px;
26.            border-radius: 12px;
27.            box-shadow: 0 10px 25px rgba(0, 0, 0, 0.08);
28.        }
29.        h1 {
30.            font-size: 22px;
31.            margin-bottom: 8px;
32.            color: #1a202c;
33.        }
34.        .subtitle {
35.            font-size: 14px;
36.            color: #718096;
37.            margin-top: 0;
38.            margin-bottom: 18px;
39.        }
```



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```
40.     textarea {
41.         width: 100%;
42.         min-height: 130px;
43.         padding: 12px;
44.         border: 1.5px solid #cbd5e0;
45.         border-radius: 6px;
46.         font-size: 14px;
47.         transition: border-color 0.2s;
48.         outline: none;
49.     }
50.     textarea:focus {
51.         border-color: #4f46e5;
52.     }
53.     .btn-submit {
54.         background-color: #4f46e5;
55.         color: #ffffff;
56.         padding: 10px 24px;
57.         border: none;
58.         border-radius: 6px;
59.         cursor: pointer;
60.         font-weight: 600;
61.         font-size: 15px;
62.         margin-top: 12px;
63.         transition: background 0.2s ease;
64.     }
65.     .btn-submit:hover {
66.         background-color: #4338ca;
67.     }
68.     .metrics-card {
69.         margin-top: 25px;
70.         border: 1px solid #e2e8f0;
71.         border-radius: 8px;
72.         background-color: #f8fafc;
73.         padding: 18px;
74.     }
75.     .metrics-card h3 {
76.         margin: 0 0 12px 0;
77.         font-size: 16px;
78.         color: #334155;
79.         border-bottom: 1px solid #e2e8f0;
80.         padding-bottom: 8px;
81.     }
82.     .stat-row {
83.         display: flex;
84.         justify-content: space-between;
85.         padding: 6px 0;
86.         font-size: 14px;
87.         border-bottom: 1px dashed #e2e8f0;
```



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```
88.     }
89.     .stat-column {
90.         display: flex;
91.         flex-direction: column;
92.         gap: 4px;
93.         padding: 6px 0;
94.         font-size: 14px;
95.         border-bottom: 1px dashed #e2e8f0;
96.     }
97.     .stat-column:last-child {
98.         border-bottom: none;
99.     }
100.    .value {
101.        font-weight: 600;
102.        color: #0f172a;
103.    }
104.    .text-preview {
105.        word-break: break-all;
106.        background: #edf2f7;
107.        padding: 8px;
108.        border-radius: 4px;
109.        font-size: 13px;
110.        color: #2d3748;
111.        max-height: 80px;
112.        overflow-y: auto;
113.    }
114.    .email-box {
115.        word-break: break-all;
116.        color: #4f46e5;
117.        font-weight: 500;
118.    }
119.    </style>
120. </head>
121. <body>
122.
123.    <div class="wrapper">
124.        <h1>Text Insights & Email Extractor</h1>
125.        <p class="subtitle">Enter any text to evaluate metrics and
126.        extract contact emails.</p>
127.        <textarea id="rawContent" placeholder="Paste your text
128.        here..."></textarea>
129.        <button class="btn-submit" id="processBtn">Process
130.        Input</button>
131.        <div class="metrics-card" id="statsBox" style="display:
132.        none;">
133.            <h3>Detailed Breakdown</h3>
```



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```
132.         <div class="stat-row">
133.             <span>Total Characters:</span>
134.             <span id="statChars" class="value">0</span>
135.         </div>
136.         <div class="stat-row">
137.             <span>Capital Letters:</span>
138.             <span id="statUpper" class="value">0</span>
139.         </div>
140.         <div class="stat-row">
141.             <span>Small Letters:</span>
142.             <span id="statLower" class="value">0</span>
143.         </div>
144.         <div class="stat-row">
145.             <span>Greeting "Welcome" Included:</span>
146.             <span id="statKeyword" class="value">No</span>
147.         </div>
148.         <div class="stat-column">
149.             <span>Uppercase Converted Text:</span>
150.             <div id="textUppercase" class="text-
preview">N/A</div>
151.         </div>
152.         <div class="stat-column">
153.             <span>Lowercase Converted Text:</span>
154.             <div id="textLowercase" class="text-
preview">N/A</div>
155.         </div>
156.         <div class="stat-column">
157.             <span>Discovered Email Addresses:</span>
158.             <span id="statEmails" class="email-box">N/A</span>
159.         </div>
160.     </div>
161. </div>
162.
163. <script>
164.     document.getElementById('processBtn').addEventListener('click',
function() {
165.         const inputData =
document.getElementById('rawContent').value;
166.
167.         // Character and letter parsing
168.         const totalChars = inputData.length;
169.         let uppercaseMatches = 0;
170.         let lowercaseMatches = 0;
171.
172.         for (let char of inputData) {
173.             if (char >= 'A' && char <= 'Z') uppercaseMatches++;
174.             else if (char >= 'a' && char <= 'z')
lowercaseMatches++;
```



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```
175.     }
176.
177.     // Keyword validation
178.     const hasWelcomeWord =
    inputData.toLowerCase().includes('welcome');
179.
180.     // Case conversions
181.     const convertedUpper = inputData.toUpperCase();
182.     const convertedLower = inputData.toLowerCase();
183.
184.     // Email extraction using pattern matching
185.     const emailPattern = /\b[A-Za-z0-9._%+-]+\@[A-Za-z0-9.-
    ]+\.[A-Za-z]{2,}\b/gi;
186.     const matchedEmails = inputData.match(emailPattern) || [];
187.     const sanitizedEmailList = Array.from(new
    Set(matchedEmails));
188.
189.     // Update DOM
190.     document.getElementById('statChars').textContent =
    totalChars;
191.     document.getElementById('statUpper').textContent =
    uppercaseMatches;
192.     document.getElementById('statLower').textContent =
    lowercaseMatches;
193.     document.getElementById('statKeyword').textContent =
    hasWelcomeWord ? "Detected" : "Not Detected";
194.     document.getElementById('textUppercase').textContent =
    inputData.trim() ? convertedUpper : "N/A";
195.     document.getElementById('textLowercase').textContent =
    inputData.trim() ? convertedLower : "N/A";
196.     document.getElementById('statEmails').textContent =
    sanitizedEmailList.length > 0 ? sanitizedEmailList.join(', ') : "No
    emails found";
197.
198.     document.getElementById('statsBox').style.display =
    'block';
199. });
200. </script>
201.
202. </body>
203. </html>
```



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4.Output:

127.0.0.1:5500/Exp6/exp6_Aaryan_Kaw...

Text Insights & Email Extractor

Enter any text to evaluate metrics and extract contact emails.

aaryankawale18@gmail.com
Welcome
Name : aaryan Kawale;
Prn:24070521162;
Sec:C ; Sem:4

Process Input

Detailed Breakdown

Total Characters:	88
Capital Letters:	7
Small Letters:	46
Greeting "Welcome" Included:	Detected
Uppercase Converted Text:	AARYANKAWALE18@GMAIL.COM WELCOME NAME : AARYAN KAWALE; PRN:24070521162; SEC:C ; SEM:4
Lowercase Converted Text:	aaryankawale18@gmail.com welcome name : aaryan kawale; prn:24070521162; sec:c ; sem:4
Discovered Email Addresses:	aaryankawale18@gmail.com



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127.0.0.1:5500/Exp6/exp6_Aaryan_Kaw...

Text Insights & Email Extractor

Enter any text to evaluate metrics and extract contact emails.

aaryankawale18@.com

Name : aaryan Kawale;
Prn:24070521162;
Sec:C ; Sem:4

Process Input

Detailed Breakdown

Total Characters:	76
Capital Letters:	6
Small Letters:	35
Greeting "Welcome" Included:	Not Detected
Uppercase Converted Text:	AARYANKAWALE18@.COM NAME : AARYAN KAWALE; PRN:24070521162; SEC:C ; SEM:4
Lowercase Converted Text:	aaryankawale18@.com name : aaryan kawale; prn:24070521162; sec:c ; sem:4
Discovered Email Addresses:	No emails found



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Case Study 6

1. **Case Study Title:** To design and implement a web application using HTML, CSS, and JavaScript to perform specific string manipulation operations (**reversing a string** and **counting vowels**) on user-provided input.
2. **Case Study Program Code:**

```
3. <!DOCTYPE html>
4. <html lang="en">
5. <head>
6.     <meta charset="UTF-8">
7.     <meta name="viewport" content="width=device-width, initial-
      scale=1.0">
8.     <title>String Operations - Single Input</title>
9.     <style>
10.         body {
11.             font-family: Arial, sans-serif;
12.             margin: 40px;
13.             background-color: #f4f6f9;
14.             color: #333;
15.         }
16.         .container {
17.             max-width: 600px;
18.             margin: 0 auto;
19.             background: #ffffff;
20.             padding: 25px;
21.             border-radius: 8px;
22.             box-shadow: 0 4px 10px rgba(0,0,0,0.1);
23.         }
24.         h2 {
25.             margin-top: 0;
26.             color: #2c3e50;
27.             border-bottom: 2px solid #ecf0f1;
28.             padding-bottom: 8px;
29.         }
30.         label {
31.             font-weight: bold;
32.             display: block;
33.             margin-bottom: 8px;
34.         }
35.         textarea {
36.             width: 100%;
37.             height: 120px;
38.             padding: 10px;
```



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```
39.         border: 1px solid #ccc;
40.         border-radius: 4px;
41.         box-sizing: border-box;
42.         font-size: 14px;
43.         resize: vertical;
44.     }
45.     .btn-group {
46.         margin-top: 15px;
47.         display: flex;
48.         gap: 10px;
49.     }
50.     button {
51.         flex: 1;
52.         padding: 10px 15px;
53.         background-color: #3498db;
54.         color: white;
55.         border: none;
56.         border-radius: 4px;
57.         cursor: pointer;
58.         font-size: 14px;
59.         font-weight: bold;
60.     }
61.     button:hover {
62.         background-color: #2980b9;
63.     }
64.     .result-box {
65.         margin-top: 20px;
66.         padding: 12px;
67.         background-color: #e8f8f5;
68.         border-left: 4px solid #1abc9c;
69.         border-radius: 2px;
70.         word-wrap: break-word;
71.         display: none;
72.     }
73. </style>
74. </head>
75. <body>
76.
77. <div class="container">
78.     <h2>String Operations</h2>
79.
80.     <label for="inputText">Enter your String / Paragraph:</label>
81.     <textarea id="inputText" placeholder="Type or paste your text
82. here..."></textarea>
83.
84.     <div class="btn-group">
85.         <button onclick="processBoth()">Run Both Operations</button>
```



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```
85.     <button onclick="reverseOnly()" style="background-color:
      #7f8c8d;">Reverse Only</button>
86.     <button onclick="countVowelsOnly()" style="background-color:
      #7f8c8d;">Count Vowels Only</button>
87.   </div>
88.
89.   <div id="outputArea" class="result-box"></div>
90.</div>
91.
92.<script>
93.  // 1. Task: Reverse the string using split()
94.  function getReversedString(text) {
95.    // split('') divides the string into an array of characters
96.    return text.split("").reverse().join("");
97.  }
98.
99.  // 2. Task: Count vowels using substring() and indexOf()
100.   function getVowelCount(text) {
101.     const vowels = "aeiouAEIOU";
102.     let vowelCount = 0;
103.
104.     for (let i = 0; i < text.length; i++) {
105.       // substring() extracts a single character
106.       let currentChar = text.substring(i, i + 1);
107.
108.       // indexOf() checks if the character is inside the
      vowel list
109.       if (vowels.indexOf(currentChar) !== -1) {
110.         vowelCount++;
111.       }
112.     }
113.     return vowelCount;
114.   }
115.
116.   // UI Handler: Run both tasks together on the single text
      input
117.   function processBoth() {
118.     const text =
      document.getElementById("inputText").value;
119.     const output = document.getElementById("outputArea");
120.
121.     if (text.trim() === "") {
122.       output.style.display = "block";
123.       output.innerHTML = "<em>Please enter some text
      first.</em>";
124.       return;
125.     }
126.
```



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```
127.         const reversed = getReversedString(text);
128.         const vowelCount = getVowelCount(text);
129.
130.         output.style.display = "block";
131.         output.innerHTML = `
132.             <strong>Reversed
133. Text:</strong><br>${reversed}<br><br>
134.             <strong>Total Vowels:</strong> ${vowelCount}
135.         `;
136.     }
137.     // Individual Handlers (optional shortcuts)
138.     function reverseOnly() {
139.         const text =
140. document.getElementById("inputText").value;
141.         const output = document.getElementById("outputArea");
142.         output.style.display = "block";
143.         output.innerHTML = `<strong>Reversed
144. Text:</strong><br>${getReversedString(text)}`;
145.     }
146.
147.     function countVowelsOnly() {
148.         const text =
149. document.getElementById("inputText").value;
150.         const output = document.getElementById("outputArea");
151.         output.style.display = "block";
152.         output.innerHTML = `<strong>Total Vowels:</strong>
153. ${getVowelCount(text)}`;
154.     }
155. </script>
156. </body>
157. </html>
```



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3.Output:

127.0.0.1:5500/Exp6/Case_Study6_Aary...

School

All Bookmarks

String Operations

Enter your String / Paragraph:

Whispers of rain danced across the tin roof as twilight slowly settled over the sleepy mountain village. Below, a narrow cobblestone path meandered through rows of centuries-old stone cottages, their windows glowing with the warm amber light of hearth fires.

Run Both Operations

Reverse Only

Count Vowels Only

Reversed Text:

.serif htreaeh fo thgil rebma mraw eht htiw gniwolg swodniw rieht ,segattoc enots dlo-seirutnec fo swor hguorht derednaem htap enotselbboc worran a ,woleB .egalliv niatnuom ypeels eht revo delttes ylwols thgiliwt sa foor nit eht ssorca decnad niar fo srepsihW

Total Vowels: 76



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Experiment No. 7

1. Experiment Title: Perform DOM Traversal and update; develop a to-do list app with add, edit, and delete features.

2. Software/Tools Required: VS Code, Html/JS Compiler...

3. Experiment Program Code:

```
4. <!DOCTYPE html>
5. <html lang="hi">
6. <head>
7.   <meta charset="UTF-8">
8.   <meta name="viewport" content="width=device-width, initial-
  scale=1.0">
9.   <title> To-Do List</title>
10.  <style>
11.    * {
12.      box-sizing: border-box;
13.      margin: 0;
14.      padding: 0;
15.      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
16.    }
17.
18.    body {
19.      min-height: 100vh;
20.      display: flex;
21.      justify-content: center;
22.      align-items: center;
23.      padding: 20px;
24.      /* Black and white aesthetic brick wall with dark gradient
  overlay */
25.      background: linear-gradient(rgba(0, 0, 0, 0.65), rgba(0, 0, 0,
    0.65)),
26.                  url('https://images.unsplash.com/photo-
    1513694203232-719a280e022f?q=80&w=1600&auto=format&fit=crop');
27.      background-size: cover;
28.      background-position: center;
29.      background-attachment: fixed;
30.    }
31.
32.    .todo-card {
33.      background: rgba(255, 255, 255, 0.92);
34.      backdrop-filter: blur(10px);
35.      -webkit-backdrop-filter: blur(10px);
36.      width: 100%;
37.      max-width: 480px;
```



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```
38.     padding: 32px 28px;
39.     border-radius: 20px;
40.     box-shadow: 0 20px 40px rgba(0, 0, 0, 0.45);
41.     border: 1px solid rgba(255, 255, 255, 0.3);
42. }
43.
44. .todo-card h2 {
45.     text-align: center;
46.     color: #1a202c;
47.     letter-spacing: 0.5px;
48.     font-size: 24px;
49.     margin-bottom: 22px;
50. }
51.
52. .input-section {
53.     display: flex;
54.     gap: 10px;
55.     margin-bottom: 22px;
56. }
57.
58. .input-section input {
59.     flex: 1;
60.     padding: 12px 16px;
61.     border: 1.5px solid #cbd5e1;
62.     border-radius: 10px;
63.     font-size: 14px;
64.     background: #ffffff;
65.     outline: none;
66.     transition: all 0.25s ease;
67. }
68.
69. .input-section input:focus {
70.     border-color: #2563eb;
71.     box-shadow: 0 0 0 3px rgba(37, 99, 235, 0.15);
72. }
73.
74. .input-section button {
75.     background: #18181b;
76.     color: #ffffff;
77.     border: none;
78.     padding: 12px 20px;
79.     border-radius: 10px;
80.     font-size: 14px;
81.     font-weight: 600;
82.     cursor: pointer;
83.     transition: all 0.2s ease;
84. }
85.
```



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```
86. .input-section button:hover {
87.     background: #27272a;
88.     transform: translateY(-1px);
89. }
90.
91. .task-list {
92.     list-style: none;
93.     max-height: 380px;
94.     overflow-y: auto;
95.     padding-right: 4px;
96. }
97.
98. .task-list::-webkit-scrollbar {
99.     width: 5px;
100. }
101.
102. .task-list::-webkit-scrollbar-thumb {
103.     background: #cbd5e1;
104.     border-radius: 10px;
105. }
106.
107. .task-item {
108.     display: flex;
109.     align-items: center;
110.     background: #ffffff;
111.     padding: 12px 14px;
112.     border-radius: 10px;
113.     margin-bottom: 10px;
114.     border: 1px solid #e2e8f0;
115.     box-shadow: 0 2px 5px rgba(0, 0, 0, 0.03);
116.     transition: transform 0.15s ease, box-shadow 0.15s ease;
117. }
118.
119. .task-item:hover {
120.     transform: translateY(-1px);
121.     box-shadow: 0 4px 10px rgba(0, 0, 0, 0.06);
122. }
123.
124. .task-item input[type="checkbox"] {
125.     width: 18px;
126.     height: 18px;
127.     cursor: pointer;
128.     margin-right: 12px;
129.     accent-color: #2563eb;
130. }
131.
132. .task-content {
133.     flex: 1;
```



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```
134.         display: flex;
135.         flex-direction: column;
136.         gap: 3px;
137.     }
138.
139.     .task-title {
140.         font-size: 14.5px;
141.         color: #1e293b;
142.         font-weight: 500;
143.         word-break: break-word;
144.     }
145.
146.     .status-tag {
147.         font-size: 11px;
148.         font-weight: 600;
149.         letter-spacing: 0.3px;
150.     }
151.
152.     .status-completed {
153.         color: #2563eb;
154.     }
155.
156.     .status-ongoing {
157.         color: #64748b;
158.     }
159.
160.     .actions {
161.         display: flex;
162.         gap: 6px;
163.         margin-left: 10px;
164.     }
165.
166.     .btn-edit, .btn-delete {
167.         border: none;
168.         padding: 6px 10px;
169.         border-radius: 6px;
170.         cursor: pointer;
171.         font-size: 12px;
172.         font-weight: 600;
173.         transition: opacity 0.2s;
174.     }
175.
176.     .btn-edit {
177.         background: #fef08a;
178.         color: #713f12;
179.     }
180.
181.     .btn-delete {
```



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```
182.         background: #fecdd3;
183.         color: #881337;
184.     }
185.
186.     .btn-edit:hover, .btn-delete:hover {
187.         opacity: 0.85;
188.     }
189. </style>
190. </head>
191. <body>
192.
193.     <div class="todo-card">
194.         <h2>Aaryan 1162</h2>
195.
196.         <div class="input-section">
197.             <input type="text" id="taskInput" placeholder="Add a new
task..." />
198.             <button onclick="addTask()">Add Task</button>
199.         </div>
200.
201.         <ul class="task-list" id="taskList"></ul>
202.     </div>
203.
204.     <script>
205.         const taskInput = document.getElementById('taskInput');
206.         const taskList = document.getElementById('taskList');
207.
208.         let tasks = JSON.parse(localStorage.getItem('my_tasks')) ||
[];
209.
210.         function saveTasks() {
211.             localStorage.setItem('my_tasks', JSON.stringify(tasks));
212.         }
213.
214.         function renderTasks() {
215.             taskList.innerHTML = '';
216.             tasks.forEach((task, index) => {
217.                 const li = document.createElement('li');
218.                 li.className = 'task-item';
219.
220.                 li.innerHTML = `
221.                     <input type="checkbox" ${task.completed ? 'checked' :
''} onchange="toggleTask(${index})">
222.                     <div class="task-content">
223.                         <span class="task-title">${task.text}</span>
224.                         <span class="status-tag ${task.completed ? 'status-
completed' : 'status-ongoing'}">
225.                             ${task.completed ? 'Completed' : 'Ongoing'}
```



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```
226.         </span>
227.     </div>
228.     <div class="actions">
229.         <button class="btn-edit"
onclick="editTask(${index})">Edit</button>
230.         <button class="btn-delete"
onclick="deleteTask(${index})">Delete</button>
231.     </div>
232.     `;
233.     taskList.appendChild(li);
234. });
235. }
236.
237. function addTask() {
238.     const text = taskInput.value.trim();
239.     if (text === '') return;
240.
241.     tasks.push({ text: text, completed: false });
242.     saveTasks();
243.     renderTasks();
244.     taskInput.value = '';
245. }
246.
247. function toggleTask(index) {
248.     tasks[index].completed = !tasks[index].completed;
249.     saveTasks();
250.     renderTasks();
251. }
252.
253. function editTask(index) {
254.     const newText = prompt('Task edit karein:',
tasks[index].text);
255.     if (newText !== null && newText.trim() !== '') {
256.         tasks[index].text = newText.trim();
257.         saveTasks();
258.         renderTasks();
259.     }
260. }
261.
262. function deleteTask(index) {
263.     tasks.splice(index, 1);
264.     saveTasks();
265.     renderTasks();
266. }
267.
268. taskInput.addEventListener('keypress', function(e) {
269.     if (e.key === 'Enter') {
270.         addTask();
```



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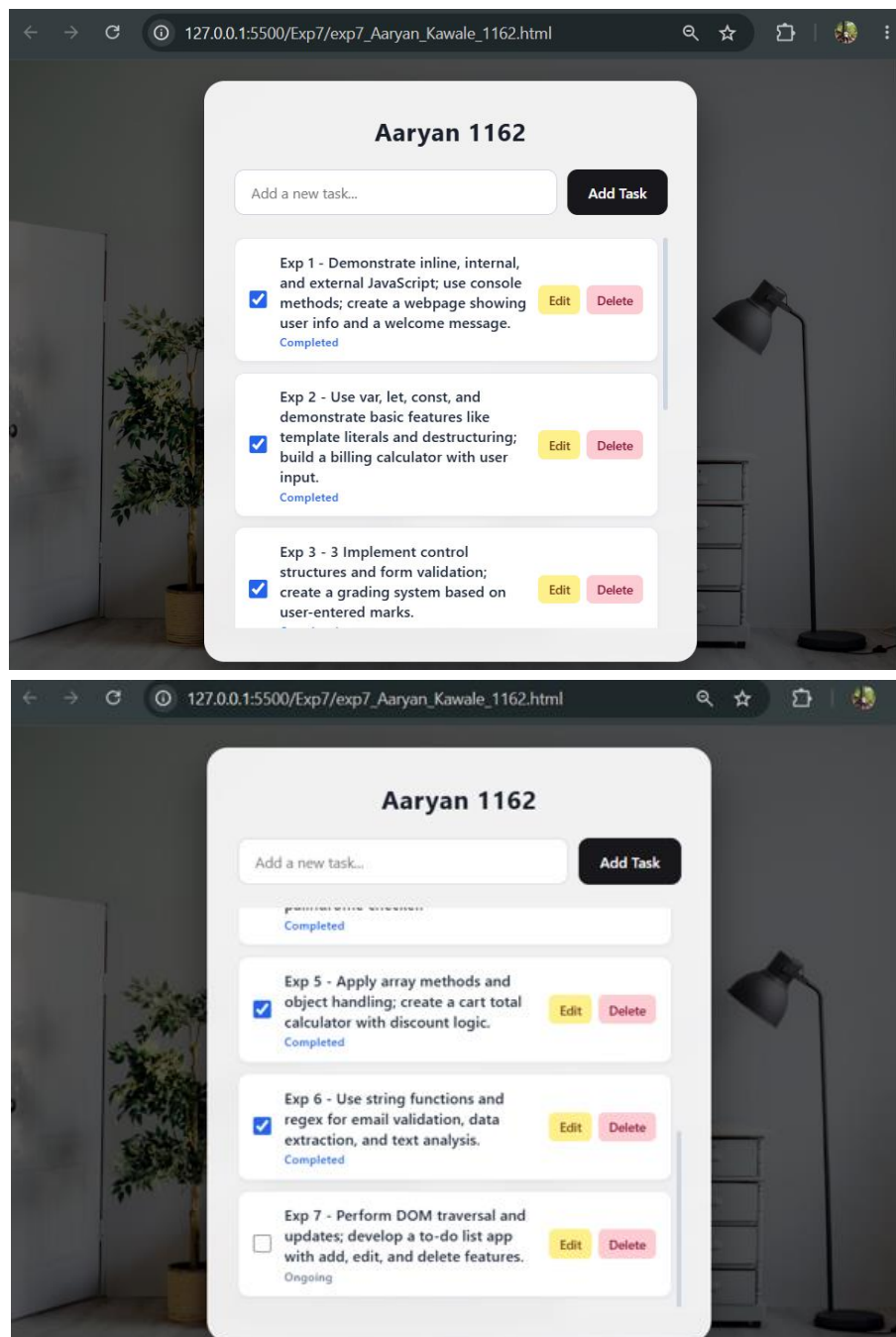
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```
271.     }  
272.     });  
273.  
274.     renderTasks();  
275. </script>  
276. </body>  
277. </html>
```

4.Output:





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Case Study 7

1. **Case Study Title:** To design, style, and implement an interactive **Student Registration Form** using **HTML5**, **CSS3**, and **JavaScript**, featuring responsive card-based layout structuring, dynamic focus/blur UI micro-interactions, and comprehensive client-side form validation.

2. Case Study Program Code:

```
3. <!DOCTYPE html>
4. <html lang="en">
5. <head>
6.     <meta charset="UTF-8">
7.     <title>Student Registration Form</title>
8.     <style>
9.         body {
10.             font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-
11. serif;
12.             background: #eef2f7;
13.             padding: 30px 15px;
14.             margin: 0;
15.         }
16.         .form-card {
17.             max-width: 580px;
18.             margin: 0 auto;
19.             background: #ffffff;
20.             padding: 30px 35px;
21.             border-radius: 10px;
22.             box-shadow: 0 4px 15px rgba(0, 0, 0, 0.08);
23.         }
24.         h2 {
25.             text-align: center;
26.             color: #2c3e50;
27.             margin-top: 0;
28.             margin-bottom: 25px;
29.             font-size: 24px;
30.         }
31.     }
32.
33.     /* 2-Column Grid to align labels side-by-side with inputs */
34.     .form-row {
35.         display: grid;
```



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```
36.         grid-template-columns: 140px 1fr;
37.         align-items: center;
38.         margin-bottom: 14px;
39.         gap: 10px;
40.     }
41.
42.     label.field-title {
43.         font-weight: 600;
44.         color: #444;
45.         font-size: 14px;
46.     }
47.
48.     .input-wrapper {
49.         display: flex;
50.         flex-direction: column;
51.     }
52.
53.     /* Inputs and Selects styling */
54.     input[type="text"],
55.     input[type="email"],
56.     input[type="date"],
57.     select {
58.         width: 100%;
59.         padding: 8px 12px;
60.         box-sizing: border-box;
61.         border: 1px solid #cbd5e1;
62.         border-radius: 5px;
63.         font-size: 14px;
64.         color: #333;
65.         transition: all 0.2s ease-in-out;
66.     }
67.
68.     /* Radio & Checkbox containers */
69.     .inline-group {
70.         display: flex;
71.         align-items: center;
72.         gap: 15px;
73.         font-size: 14px;
74.         color: #444;
75.     }
76.
77.     .inline-group label {
78.         display: inline-flex;
79.         align-items: center;
80.         gap: 5px;
81.         cursor: pointer;
82.         font-weight: normal;
83.     }
```



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```
84.
85.     .error {
86.         color: #e74c3c;
87.         font-size: 12px;
88.         margin-top: 3px;
89.     }
90.
91.     /* Submit Button */
92.     .submit-btn {
93.         width: 100%;
94.         padding: 11px;
95.         background-color: #3498db;
96.         color: #ffffff;
97.         border: none;
98.         border-radius: 5px;
99.         font-size: 15px;
100.        font-weight: 600;
101.        cursor: pointer;
102.        transition: background-color 0.2s;
103.        margin-top: 10px;
104.    }
105.
106.    .submit-btn:hover {
107.        background-color: #2980b9;
108.    }
109.
110.    /* Submission message display */
111.    #successMessage {
112.        display: none;
113.        margin-top: 20px;
114.        padding: 12px;
115.        background-color: #d4edda;
116.        color: #155724;
117.        border: 1px solid #c3e6cb;
118.        border-radius: 5px;
119.        text-align: center;
120.        font-weight: 600;
121.        font-size: 15px;
122.    }
123.    </style>
124. </head>
125. <body>
126.
127. <div class="form-card">
128.     <h2>Student Registration</h2>
129.
130.     <form onsubmit="return validateForm(event)">
131.
```



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```
132.         <!-- Name -->
133.         <div class="form-row">
134.             <label class="field-title" for="name">Name:</label>
135.             <div class="input-wrapper">
136.                 <input type="text" id="name" placeholder="Full
name" onfocus="highlight(this)" onblur="unhighlight(this)">
137.                 <span id="nameError" class="error"></span>
138.             </div>
139.         </div>
140.
141.         <!-- Email -->
142.         <div class="form-row">
143.             <label class="field-title"
for="email">Email:</label>
144.             <div class="input-wrapper">
145.                 <input type="email" id="email"
placeholder="example@email.com" onfocus="highlight(this)"
onblur="unhighlight(this)">
146.                 <span id="emailError" class="error"></span>
147.             </div>
148.         </div>
149.
150.         <!-- Date of Birth -->
151.         <div class="form-row">
152.             <label class="field-title" for="dob">Date of
Birth:</label>
153.             <div class="input-wrapper">
154.                 <input type="date" id="dob"
onfocus="highlight(this)" onblur="unhighlight(this)">
155.                 <span id="dobError" class="error"></span>
156.             </div>
157.         </div>
158.
159.         <!-- Gender -->
160.         <div class="form-row">
161.             <label class="field-title">Gender:</label>
162.             <div class="input-wrapper">
163.                 <div class="inline-group">
164.                     <label><input type="radio" name="gender"
value="Male" onchange="clearError('genderError')"> Male</label>
165.                     <label><input type="radio" name="gender"
value="Female" onchange="clearError('genderError')"> Female</label>
166.                     <label><input type="radio" name="gender"
value="Other" onchange="clearError('genderError')"> Other</label>
167.                 </div>
168.                 <span id="genderError" class="error"></span>
169.             </div>
170.         </div>
```



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```
171.
172.         <!-- Year -->
173.         <div class="form-row">
174.             <label class="field-title" for="year">Year:</label>
175.             <div class="input-wrapper">
176.                 <select id="year" onfocus="highlight(this)"
onblur="unhighlight(this)" onchange="clearError('yearError')">
177.                     <option value="">-- Select Year --</option>
178.                     <option value="1st Year">1st Year</option>
179.                     <option value="2nd Year">2nd Year</option>
180.                     <option value="3rd Year">3rd Year</option>
181.                     <option value="4th Year">4th Year</option>
182.                 </select>
183.                 <span id="yearError" class="error"></span>
184.             </div>
185.         </div>
186.
187.         <!-- Semester -->
188.         <div class="form-row">
189.             <label class="field-title"
for="sem">Semester:</label>
190.             <div class="input-wrapper">
191.                 <select id="sem" onfocus="highlight(this)"
onblur="unhighlight(this)" onchange="clearError('semError')">
192.                     <option value="">-- Select Semester --
</option>
193.                     <option value="1">Semester 1</option>
194.                     <option value="2">Semester 2</option>
195.                     <option value="3">Semester 3</option>
196.                     <option value="4">Semester 4</option>
197.                     <option value="5">Semester 5</option>
198.                     <option value="6">Semester 6</option>
199.                     <option value="7">Semester 7</option>
200.                     <option value="8">Semester 8</option>
201.                 </select>
202.                 <span id="semError" class="error"></span>
203.             </div>
204.         </div>
205.
206.         <!-- Section -->
207.         <div class="form-row">
208.             <label class="field-title"
for="sec">Section:</label>
209.             <div class="input-wrapper">
210.                 <input type="text" id="sec" placeholder="e.g.,
A, B, C" onfocus="highlight(this)" onblur="unhighlight(this)">
211.                 <span id="secError" class="error"></span>
212.             </div>
```



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```
213.         </div>
214.
215.         <!-- Batch -->
216.         <div class="form-row">
217.             <label class="field-title"
for="batch">Batch:</label>
218.             <div class="input-wrapper">
219.                 <input type="text" id="batch"
placeholder="e.g., 2023-2027" onfocus="highlight(this)"
onblur="unhighlight(this)">
220.                 <span id="batchError" class="error"></span>
221.             </div>
222.         </div>
223.
224.         <!-- Terms and Conditions -->
225.         <div class="form-row">
226.             <div></div>
227.             <div class="input-wrapper">
228.                 <div class="inline-group">
229.                     <label>
230.                         <input type="checkbox" id="terms"
onchange="clearError('termsError')">
231.                         I agree to all terms and conditions
232.                     </label>
233.                 </div>
234.                 <span id="termsError" class="error"></span>
235.             </div>
236.         </div>
237.
238.         <!-- Submit Button -->
239.         <input type="submit" class="submit-btn" value="Submit
Form">
240.     </form>
241.
242.     <div id="successMessage">Form submitted successfully!</div>
243. </div>
244.
245. <script>
246.     // Focus Event: Light blue highlight with subtle border
247.     function highlight(x) {
248.         x.style.backgroundColor = "#f0f7ff";
249.         x.style.borderColor = "#3498db";
250.         x.style.outline = "none";
251.     }
252.
253.     // Blur Event: Revert styling
254.     function unhighlight(x) {
255.         x.style.backgroundColor = "#ffffff";
```



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```
256.         x.style.borderColor = "#cbd5e1";
257.     }
258.
259.     // Change Event Helper: Clears specific error message
260.     function clearError(errorId) {
261.         document.getElementById(errorId).innerText = "";
262.     }
263.
264.     // Submit Event & Validations
265.     function validateForm(event) {
266.         event.preventDefault(); // Stop default form submit
267.
268.         // Clear all previous error messages
269.         var errors = document.querySelectorAll(".error");
270.         errors.forEach(function(el) { el.innerText = ""; });
271.         document.getElementById("successMessage").style.display
= "none";
272.
273.         // Read values
274.         var name =
document.getElementById("name").value.trim();
275.         var email =
document.getElementById("email").value.trim();
276.         var dob = document.getElementById("dob").value;
277.         var year = document.getElementById("year").value;
278.         var sem = document.getElementById("sem").value;
279.         var sec = document.getElementById("sec").value.trim();
280.         var batch =
document.getElementById("batch").value.trim();
281.         var terms = document.getElementById("terms").checked;
282.
283.         // Check gender radio button
284.         var gender = "";
285.         var radios = document.getElementsByName("gender");
286.         for (var i = 0; i < radios.length; i++) {
287.             if (radios[i].checked) {
288.                 gender = radios[i].value;
289.                 break;
290.             }
291.         }
292.
293.         var isValid = true;
294.
295.         if (name === "") {
296.             document.getElementById("nameError").innerText =
"Name is required";
297.             isValid = false;
298.         }
```



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```
299.
300.         if (email === "") {
301.             document.getElementById("emailError").innerText =
"Email is required";
302.             isValid = false;
303.         } else if (email.indexOf("@") === -1 ||
email.indexOf(".") === -1) {
304.             document.getElementById("emailError").innerText =
"Valid email required (@ and .)";
305.             isValid = false;
306.         }
307.
308.         if (dob === "") {
309.             document.getElementById("dobError").innerText =
"Date of Birth is required";
310.             isValid = false;
311.         }
312.
313.         if (gender === "") {
314.             document.getElementById("genderError").innerText =
"Please select gender";
315.             isValid = false;
316.         }
317.
318.         if (year === "") {
319.             document.getElementById("yearError").innerText =
"Please select year";
320.             isValid = false;
321.         }
322.
323.         if (sem === "") {
324.             document.getElementById("semError").innerText =
"Please select semester";
325.             isValid = false;
326.         }
327.
328.         if (sec === "") {
329.             document.getElementById("secError").innerText =
"Section is required";
330.             isValid = false;
331.         }
332.
333.         if (batch === "") {
334.             document.getElementById("batchError").innerText =
"Batch is required";
335.             isValid = false;
336.         }
337.
```



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```
338.         if (!terms) {
339.             document.getElementById("termsError").innerText =
    "You must agree to the terms";
340.             isValid = false;
341.         }
342.
343.         // Show success notification if all checks pass
344.         if (isValid) {
345.             var msg =
    document.getElementById("successMessage");
346.             msg.style.display = "block";
347.             msg.innerText = "Form submitted successfully!";
348.         }
349.
350.         return false;
351.     }
352. </script>
353.
354. </body>
355. </html>
```

3.Output:

Student Registration

Name: Aaryan Kawale

Email: aaryankawale18@gmail.com

Date of Birth: 12/09/2006

Gender: ☒ Male ☐ Female ☐ Other

Year: 3rd Year

Semester: Semester 5

Section: A

Batch: 2024-2028

☒ I agree to all terms and conditions

Submit Form

Form submitted successfully!



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127.0.0.1:5500/Exp7/Case_Study7_Aaryan_Kawale_1162.html

All Bookmarks

Student Registration

Name:

Aaryan Kawale

Email:

aaryankawale18mail.com

Date of Birth:

1

Gender:

☒ Male ☐ Female ☐ Other

Year:

3rd Year

Semester:

Semester 5

Section:

A

Batch:

2024-2028

☒ I agree to all terms and conditions

Submit Form

Form submitted successfully!



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Experiment No. 8

1. Experiment Title: Access and validate form fields using events; design a gym registration form with live input checks.

2. Software/Tools Required: VS Code, Html/JS Compiler...

3. Experiment Program Code:

```
4. <!DOCTYPE html>
5. <html lang="en">
6. <head>
7.   <meta charset="UTF-8" />
8.   <meta name="viewport" content="width=device-width, initial-
  scale=1.0" />
9.   <title>Gym Membership Registration</title>
10.  <style>
11.    * {
12.      box-sizing: border-box;
13.      margin: 0;
14.      padding: 0;
15.      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
16.    }
17.
18.    body {
19.      background: linear-gradient(135deg, #1f1c2c, #928dab);
20.      min-height: 100vh;
21.      display: flex;
22.      justify-content: center;
23.      align-items: center;
24.      padding: 24px;
25.    }
26.
27.    .form-card {
28.      background: #ffffff;
29.      border-radius: 12px;
30.      padding: 32px;
31.      width: 100%;
32.      max-width: 520px;
33.      box-shadow: 0 10px 25px rgba(0, 0, 0, 0.25);
34.    }
35.
36.    .form-card h2 {
37.      text-align: center;
38.      margin-bottom: 8px;
39.      color: #222;
40.    }
```



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```
41.
42.   .form-card p.subtitle {
43.     text-align: center;
44.     font-size: 14px;
45.     color: #666;
46.     margin-bottom: 24px;
47.   }
48.
49.   .form-group {
50.     margin-bottom: 16px;
51.   }
52.
53.   label {
54.     display: block;
55.     margin-bottom: 6px;
56.     font-size: 14px;
57.     font-weight: 600;
58.     color: #333;
59.   }
60.
61.   input, select {
62.     width: 100%;
63.     padding: 10px 12px;
64.     border: 1px solid #ccc;
65.     border-radius: 6px;
66.     font-size: 14px;
67.     outline: none;
68.     transition: border-color 0.2s;
69.   }
70.
71.   input:focus, select:focus {
72.     border-color: #ff5722;
73.   }
74.
75.   input.input-error, select.input-error {
76.     border-color: #e53935;
77.   }
78.
79.   .error-msg {
80.     display: block;
81.     color: #e53935;
82.     font-size: 12px;
83.     margin-top: 4px;
84.     min-height: 16px;
85.   }
86.
87.   .row {
88.     display: flex;
```



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```
89.     gap: 12px;
90.   }
91.
92.   .row .form-group {
93.     flex: 1;
94.   }
95.
96.   button[type="submit"] {
97.     width: 100%;
98.     padding: 12px;
99.     background-color: #ff5722;
100.    color: #ffffff;
101.    border: none;
102.    border-radius: 6px;
103.    font-size: 16px;
104.    font-weight: bold;
105.    cursor: pointer;
106.    transition: background-color 0.2s;
107.    margin-top: 8px;
108.  }
109.
110.  button[type="submit"]:hover {
111.    background-color: #e64a19;
112.  }
113.
114.  .confirmation-box {
115.    display: none;
116.    margin-top: 20px;
117.    padding: 14px;
118.    background-color: #e8f5e9;
119.    border: 1px solid #a5d6a7;
120.    color: #2e7d32;
121.    border-radius: 6px;
122.    text-align: center;
123.    font-size: 15px;
124.    font-weight: 600;
125.  }
126.  </style>
127. </head>
128. <body>
129.
130.  <div class="form-card">
131.    <h2>FitZone Gym</h2>
132.    <p class="subtitle">Member Registration Form</p>
133.
134.    <form id="gymForm" novalidate>
135.      <div class="form-group">
136.        <label for="fullName">Full Name</label>
```



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```
137.         <input type="text" id="fullName" placeholder="Enter
your full name" />
138.         <span class="error-msg" id="nameError"></span>
139.     </div>
140.
141.     <div class="row">
142.         <div class="form-group">
143.             <label for="age">Age</label>
144.             <input type="number" id="age" placeholder="Age (15+)"
/>
145.             <span class="error-msg" id="ageError"></span>
146.         </div>
147.         <div class="form-group">
148.             <label for="gender">Gender</label>
149.             <select id="gender">
150.                 <option value="">Select Gender</option>
151.                 <option value="Male">Male</option>
152.                 <option value="Female">Female</option>
153.                 <option value="Other">Other</option>
154.             </select>
155.             <span class="error-msg" id="genderError"></span>
156.         </div>
157.     </div>
158.
159.     <div class="form-group">
160.         <label for="email">Email Address</label>
161.         <input type="email" id="email"
placeholder="example@domain.com" />
162.         <span class="error-msg" id="emailError"></span>
163.     </div>
164.
165.     <div class="form-group">
166.         <label for="phone">Phone Number</label>
167.         <input type="tel" id="phone" placeholder="10-digit
mobile number" />
168.         <span class="error-msg" id="phoneError"></span>
169.     </div>
170.
171.     <div class="form-group">
172.         <label for="membership">Membership Type</label>
173.         <select id="membership">
174.             <option value="">Select a Plan</option>
175.             <option value="Standard Monthly">Standard Monthly
(4000/mo)</option>
176.             <option value="Quarterly Pass">Quarterly Pass (8500/3
mos)</option>
177.             <option value="Annual Premium">Annual Premium
(24999/yr)</option>
```



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```
178.         <option value="VIP Personal Training">VIP All-Access
+ Trainer (30500/yr)</option>
179.         </select>
180.         <span class="error-msg" id="membershipError"></span>
181.     </div>
182.
183.     <button type="submit" id="submitBtn">Register
Now</button>
184. </form>
185.
186.     <div class="confirmation-box" id="confirmationBox">
187.         Registration submitted successfully! Welcome to the gym.
188.     </div>
189. </div>
190.
191.     <script>
192.         const form = document.getElementById('gymForm');
193.         const confirmationBox =
document.getElementById('confirmationBox');
194.
195.         function showError(inputId, errorId, message) {
196.             const input = document.getElementById(inputId);
197.             const error = document.getElementById(errorId);
198.             input.classList.add('input-error');
199.             error.textContent = message;
200.         }
201.
202.         function clearError(inputId, errorId) {
203.             const input = document.getElementById(inputId);
204.             const error = document.getElementById(errorId);
205.             input.classList.remove('input-error');
206.             error.textContent = '';
207.         }
208.
209.         form.addEventListener('submit', function(e) {
210.             e.preventDefault();
211.             confirmationBox.style.display = 'none';
212.
213.             let isValid = true;
214.
215.             const name =
document.getElementById('fullName').value.trim();
216.             const age =
parseInt(document.getElementById('age').value, 10);
217.             const gender = document.getElementById('gender').value;
218.             const email =
document.getElementById('email').value.trim();
```



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```
219.         const phone =
document.getElementById('phone').value.trim();
220.         const membership =
document.getElementById('membership').value;
221.
222.         // Name validation
223.         if (!name) {
224.             showError('fullName', 'nameError', 'Please enter your
full name.');
```

```
225.             isValid = false;
226.         } else {
227.             clearError('fullName', 'nameError');
```

```
228.         }
229.
230.         // Age validation (15+)
231.         if (isNaN(age)) {
232.             showError('age', 'ageError', 'Please enter your age.');
```

```
233.             isValid = false;
234.         } else if (age < 15) {
235.             showError('age', 'ageError', 'Age must be 15 or older
to register.');
```

```
236.             isValid = false;
237.         } else {
238.             clearError('age', 'ageError');
```

```
239.         }
240.
241.         // Gender validation
242.         if (!gender) {
243.             showError('gender', 'genderError', 'Please select a
gender.');
```

```
244.             isValid = false;
245.         } else {
246.             clearError('gender', 'genderError');
```

```
247.         }
248.
249.         // Email validation
250.         const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
```

```
251.         if (!email) {
252.             showError('email', 'emailError', 'Please enter your
email address.');
```

```
253.             isValid = false;
254.         } else if (!emailRegex.test(email)) {
255.             showError('email', 'emailError', 'Invalid email address
format.');
```

```
256.             isValid = false;
257.         } else {
258.             clearError('email', 'emailError');
```

```
259.         }
```



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```
260.
261.     // Phone validation (10 digits)
262.     const phoneDigits = phone.replace(/\D/g, '');
263.     if (!phone) {
264.         showError('phone', 'phoneError', 'Please enter your
phone number.');
```

```
265.         isValid = false;
266.     } else if (phoneDigits.length !== 10) {
267.         showError('phone', 'phoneError', 'Please enter a valid
10-digit phone number.');
```

```
268.         isValid = false;
269.     } else {
270.         clearError('phone', 'phoneError');
```

```
271.     }
272.
273.     // Membership validation
274.     if (!membership) {
275.         showError('membership', 'membershipError', 'Please
select a membership plan.');
```

```
276.         isValid = false;
277.     } else {
278.         clearError('membership', 'membershipError');
```

```
279.     }
280.
281.     // Final check
282.     if (isValid) {
283.         confirmationBox.style.display = 'block';
284.         form.reset();
285.     }
286. });
287. </script>
288. </body>
289. </html>
```



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4.Output:

127.0.0.1:5500/Exp8/exp8_Aaryan_Kawale_1162.html

FitZone Gym

Member Registration Form

Full Name

Aaryan Kawale

Age

20

Gender

Male

Email Address

aaryankawale18@gmail.com

Phone Number

9420910521

Membership Type

Standard Monthly (4000/mo)

Register Now

Registration submitted successfully! Welcome to the gym.

Mouza - Wathoda, Nagpur 440008, Maharashtra, India. | Tel: +91-712-6192370 / +91-712-6192368
Email: info@sitnagpur.siu.edu.in | Website: www.sitnagpur.edu.in



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127.0.0.1:5500/Exp8/exp8_Aaryan_Kawale_1162.html

FitZone Gym

Member Registration Form

Full Name

Aaryan Kawale

Age

14

Gender

Male

Age must be 15 or older to register.

Email Address

aaryankawale18+@.com

Invalid email address format.

Phone Number

24070521162

Please enter a valid 10-digit phone number.

Membership Type

Quarterly Pass (8500/3 mos)

Register Now



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Case Study 8

1. Case Study Title: To design and implement a responsive, multi-section **Job Application Form** styled with modern CSS gradients and card layout.

2. Case Study Program Code:

```
3. <!DOCTYPE html>
4. <html lang="en">
5. <head>
6. <meta charset="UTF-8">
7. <meta name="viewport" content="width=device-width, initial-scale=1.0">
8.
9. <title>Job Application</title>
10.
11. <style>
12. * {
13. margin: 0;
14. padding: 0;
15. box-sizing: border-box;
16. font-family: "Segoe UI", Arial, sans-serif;
17. }
18.
19. body {
20. min-height: 100vh;
21. background: linear-gradient(135deg, #141e30, #243b55);
22. padding: 35px 15px;
23. color: #222;
24. }
25.
26. /* MAIN CARD */
27.
28. .container {
29. max-width: 1050px;
30. margin: auto;
31. background: #ffffff;
32. border-radius: 22px;
33. overflow: hidden;
34. box-shadow: 0 20px 60px rgba(0,0,0,0.25);
35. }
36.
37. /* HEADER */
38.
39. .header {
40. background: linear-gradient(135deg, #6a11cb, #2575fc);
41. color: white;
42. padding: 45px 40px;
```



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```
43.position: relative;
44.}
45.
46..header h1 {
47.font-size: 34px;
48.margin-bottom: 8px;
49.}
50.
51..header p {
52.opacity: 0.9;
53.font-size: 15px;
54.}
55.
56..header-badge {
57.position: absolute;
58.right: 40px;
59.top: 45px;
60.background: rgba(255,255,255,0.18);
61.padding: 12px 18px;
62.border-radius: 30px;
63.font-size: 13px;
64.}
65.
66./* FORM */
67.
68..form-container {
69.padding: 40px;
70.}
71.
72./* SECTION */
73.
74..section {
75.margin-bottom: 42px;
76.}
77.
78..section-heading {
79.display: flex;
80.align-items: center;
81.gap: 15px;
82.margin-bottom: 25px;
83.}
84.
85..number {
86.width: 40px;
87.height: 40px;
88.border-radius: 50%;
89.background: linear-gradient(135deg, #6a11cb, #2575fc);
90.color: white;
```



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```
91.display: flex;
92.justify-content: center;
93.align-items: center;
94.font-weight: bold;
95.}
96.
97..section-heading h2 {
98.font-size: 21px;
99.color: #222;
100.    }
101.
102.    .section-heading p {
103.        font-size: 12px;
104.        color: #888;
105.        margin-top: 3px;
106.    }
107.
108.    /* GRID */
109.
110.    .form-grid {
111.        display: grid;
112.        grid-template-columns: 1fr 1fr;
113.        gap: 22px;
114.    }
115.
116.    .full {
117.        grid-column: 1 / -1;
118.    }
119.
120.    /* INPUT */
121.
122.    .form-group {
123.        position: relative;
124.    }
125.
126.    label {
127.        display: block;
128.        font-size: 13px;
129.        font-weight: 600;
130.        color: #555;
131.        margin-bottom: 8px;
132.    }
133.
134.    .required {
135.        color: #e63946;
136.    }
137.
138.    input,
```



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```
139.     select,
140.     textarea {
141.         width: 100%;
142.         border: 1.5px solid #ddd;
143.         background: #fafafa;
144.         border-radius: 10px;
145.         padding: 13px 15px;
146.         font-size: 14px;
147.         outline: none;
148.         transition: 0.25s;
149.     }
150.
151.     input:focus,
152.     select:focus,
153.     textarea:focus {
154.         background: white;
155.         border-color: #6a11cb;
156.         box-shadow: 0 0 0 4px rgba(106,17,203,0.08);
157.     }
158.
159.     textarea {
160.         resize: vertical;
161.         min-height: 100px;
162.     }
163.
164.     /* VALID */
165.
166.     input.valid,
167.     select.valid,
168.     textarea.valid {
169.         border-color: #20a464;
170.         background: #f6fff9;
171.     }
172.
173.     /* INVALID */
174.
175.     input.invalid,
176.     select.invalid,
177.     textarea.invalid {
178.         border-color: #e63946;
179.         background: #fff7f7;
180.     }
181.
182.     /* ERROR */
183.
184.     .error {
185.         min-height: 18px;
186.         margin-top: 5px;
```



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```
187.     font-size: 12px;
188.     color: #e63946;
189.     }
190.
191.     /* VALID ICON */
192.
193.     .valid-icon {
194.     position: absolute;
195.     right: 13px;
196.     top: 38px;
197.     color: #20a464;
198.     font-weight: bold;
199.     display: none;
200.     }
201.
202.     .form-group.is-valid .valid-icon {
203.     display: block;
204.     }
205.
206.     /* GENDER */
207.
208.     .gender-container {
209.     display: flex;
210.     gap: 12px;
211.     flex-wrap: wrap;
212.     }
213.
214.     .gender-option {
215.     position: relative;
216.     }
217.
218.     .gender-option input {
219.     display: none;
220.     }
221.
222.     .gender-option label {
223.     margin: 0;
224.     padding: 11px 20px;
225.     border: 1.5px solid #ddd;
226.     border-radius: 30px;
227.     cursor: pointer;
228.     background: #fafafa;
229.     transition: 0.2s;
230.     }
231.
232.     .gender-option input:checked + label {
233.     background: linear-gradient(135deg, #6a11cb, #2575fc);
234.     color: white;
```



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```
235.     border-color: transparent;
236.   }
237.
238.   /* SKILLS */
239.
240.   .skill-box {
241.     display: flex;
242.     flex-wrap: wrap;
243.     gap: 8px;
244.     margin-top: 8px;
245.   }
246.
247.   .skill {
248.     background: #eee8ff;
249.     color: #5b16b5;
250.     padding: 6px 12px;
251.     border-radius: 20px;
252.     font-size: 12px;
253.   }
254.
255.   /* BUTTONS */
256.
257.   .buttons {
258.     border-top: 1px solid #eee;
259.     padding-top: 30px;
260.     display: flex;
261.     justify-content: flex-end;
262.     gap: 12px;
263.   }
264.
265.   button {
266.     border: none;
267.     border-radius: 10px;
268.     padding: 14px 25px;
269.     font-size: 14px;
270.     font-weight: 600;
271.     cursor: pointer;
272.     transition: 0.2s;
273.   }
274.
275.   .submit-btn {
276.     color: white;
277.     background: linear-gradient(135deg, #6a11cb, #2575fc);
278.   }
279.
280.   .submit-btn:hover {
281.     transform: translateY(-2px);
282.     box-shadow: 0 8px 20px rgba(37,117,252,0.3);
```



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```
283.     }
284.
285.     .reset-btn {
286.     background: #eee;
287.     color: #444;
288.     }
289.
290.     .reset-btn:hover {
291.     background: #ddd;
292.     }
293.
294.     /* SUCCESS */
295.
296.     .success {
297.     display: none;
298.     margin-top: 25px;
299.     padding: 18px;
300.     background: #e9fff2;
301.     border: 1px solid #9de7bc;
302.     color: #147a43;
303.     border-radius: 10px;
304.     text-align: center;
305.     font-weight: 600;
306.     }
307.
308.     /* FOOTER */
309.
310.     .footer {
311.     text-align: center;
312.     color: #999;
313.     font-size: 12px;
314.     padding: 0 40px 30px;
315.     }
316.
317.     /* MOBILE */
318.
319.     @media(max-width: 700px) {
320.
321.     body {
322.     padding: 10px;
323.     }
324.
325.     .header {
326.     padding: 30px 22px;
327.     }
328.
329.     .header h1 {
330.     font-size: 25px;
```



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```
331.     }
332.
333.     .header-badge {
334.     display: none;
335.     }
336.
337.     .form-container {
338.     padding: 25px 20px;
339.     }
340.
341.     .form-grid {
342.     grid-template-columns: 1fr;
343.     }
344.
345.     .full {
346.     grid-column: auto;
347.     }
348.
349.     .section-heading h2 {
350.     font-size: 18px;
351.     }
352.
353.     .buttons {
354.     flex-direction: column;
355.     }
356.
357.     button {
358.     width: 100%;
359.     }
360.     }
361. </style>
362. </head>
363.
364. <body>
365.
366. <div class="container">
367.
368. <!-- HEADER -->
369.
370. <div class="header">
371.
372. <div class="header-badge">
373. Online Application
374. </div>
375.
376. <h1>Job Application</h1>
377.
```



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```
378.      <p>
379.      Take the next step in your career. Fill in your details below.
380.      </p>
381.
382.      </div>
383.
384.      <div class="form-container">
385.
386.      <form id="jobForm" novalidate>
387.
388.      <!-- ===== -->
389.      <!-- PERSONAL INFORMATION -->
390.      <!-- ===== -->
391.
392.      <div class="section">
393.
394.      <div class="section-heading">
395.
396.      <div class="number">01</div>
397.
398.      <div>
399.      <h2>Personal Information</h2>
400.      <p>Tell us about yourself</p>
401.      </div>
402.
403.      </div>
404.
405.      <div class="form-grid">
406.
407.      <!-- FULL NAME -->
408.
409.      <div class="form-group full">
410.
411.      <label>
412.      Full Name <span class="required">*</span>
413.      </label>
414.
415.      <input
416.      type="text"
417.      id="fullName"
418.      placeholder="Enter your full name"
419.      >
420.
421.      <span class="valid-icon">✓</span>
422.
```



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```
423.     <div class="error" id="fullNameError"></div>
424.
425.     </div>
426.
427.     <!-- DOB -->
428.
429.     <div class="form-group">
430.
431.         <label>
432.             Date of Birth <span class="required">*</span>
433.         </label>
434.
435.         <input
436.             type="date"
437.             id="dob"
438.         >
439.
440.     <div class="error" id="dobError"></div>
441.
442.     </div>
443.
444.     <!-- GENDER -->
445.
446.     <div class="form-group">
447.
448.         <label>
449.             Gender <span class="required">*</span>
450.         </label>
451.
452.         <div class="gender-container">
453.
454.             <div class="gender-option">
455.
456.                 <input
457.                     type="radio"
458.                     name="gender"
459.                     id="male"
460.                     value="Male"
461.                 >
462.
463.                 <label for="male">Male</label>
464.
465.             </div>
466.
467.             <div class="gender-option">
468.
```



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```
469.     <input
470.     type="radio"
471.     name="gender"
472.     id="female"
473.     value="Female"
474.     >
475.
476.     <label for="female">Female</label>
477.
478. </div>
479.
480. <div class="gender-option">
481.
482.     <input
483.     type="radio"
484.     name="gender"
485.     id="other"
486.     value="Other"
487.     >
488.
489.     <label for="other">Other</label>
490.
491. </div>
492.
493. </div>
494.
495. <div class="error" id="genderError"></div>
496.
497. </div>
498.
499. <!-- EMAIL -->
500.
501. <div class="form-group">
502.
503.     <label>
504.     Email Address <span class="required">*</span>
505.     </label>
506.
507.     <input
508.     type="text"
509.     id="email"
510.     placeholder="example@gmail.com"
511.     >
512.
513.     <span class="valid-icon">✓</span>
514.
```



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```
515.     <div class="error" id="emailError"></div>
516.
517.     </div>
518.
519.     <!-- MOBILE -->
520.
521.     <div class="form-group">
522.
523.         <label>
524.             Mobile Number <span class="required">*</span>
525.         </label>
526.
527.         <input
528.             type="text"
529.             id="mobile"
530.             maxlength="10"
531.             placeholder="10 digit mobile number"
532.         >
533.
534.         <span class="valid-icon">✓</span>
535.
536.     <div class="error" id="mobileError"></div>
537.
538. </div>
539.
540.     <!-- ADDRESS -->
541.
542.     <div class="form-group full">
543.
544.         <label>
545.             Address <span class="required">*</span>
546.         </label>
547.
548.         <textarea
549.             id="address"
550.             placeholder="Enter your complete address"
551.         ></textarea>
552.
553.     <div class="error" id="addressError"></div>
554.
555. </div>
556.
557.     <!-- CITY -->
558.
559.     <div class="form-group">
560.
```



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```
561.     <label>
562.     City <span class="required">*</span>
563.     </label>
564.
565.     <input
566.     type="text"
567.     id="city"
568.     placeholder="Enter city"
569.     >
570.
571.     <span class="valid-icon">✓</span>
572.
573.     <div class="error" id="cityError"></div>
574.
575.     </div>
576.
577.     <!-- STATE -->
578.
579.     <div class="form-group">
580.
581.     <label>
582.     State <span class="required">*</span>
583.     </label>
584.
585.     <input
586.     type="text"
587.     id="state"
588.     placeholder="Enter state"
589.     >
590.
591.     <span class="valid-icon">✓</span>
592.
593.     <div class="error" id="stateError"></div>
594.
595.     </div>
596.
597.     <!-- PIN -->
598.
599.     <div class="form-group">
600.
601.     <label>
602.     PIN Code <span class="required">*</span>
603.     </label>
604.
605.     <input
606.     type="text"
```



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```
607.     id="pin"
608.     maxlength="6"
609.     placeholder="6 digit PIN"
610.     >
611.
612.     <span class="valid-icon">✓</span>
613.
614.     <div class="error" id="pinError"></div>
615.
616. </div>
617.
618. </div>
619. </div>
620.
621. <!-- ===== -->
622. <!-- EDUCATION -->
623. <!-- ===== -->
624.
625. <div class="section">
626.
627.     <div class="section-heading">
628.
629.         <div class="number">02</div>
630.
631.         <div>
632.             <h2>Educational Qualification</h2>
633.             <p>Your academic background</p>
634.         </div>
635.
636.     </div>
637.
638.     <div class="form-grid">
639.
640.         <div class="form-group">
641.
642.             <label>
643.                 Highest Qualification <span class="required">*</span>
644.             </label>
645.
646.             <select id="qualification">
647.
648.                 <option value="">
649.                     Select qualification
650.                 </option>
651.
652.                 <option>10th</option>
```



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```
653.      <option>12th</option>
654.      <option>Diploma</option>
655.      <option>Bachelor's Degree</option>
656.      <option>Master's Degree</option>
657.      <option>PhD</option>
658.
659.      </select>
660.
661.      <div class="error" id="qualificationError"></div>
662.
663.      </div>
664.
665.      <div class="form-group">
666.
667.      <label>
668.      University / Institute <span class="required">*</span>
669.      </label>
670.
671.      <input
672.      type="text"
673.      id="university"
674.      placeholder="University / Institute name"
675.      >
676.
677.      <span class="valid-icon">✓</span>
678.
679.      <div class="error" id="universityError"></div>
680.
681.      </div>
682.
683.      <div class="form-group">
684.
685.      <label>
686.      Passing Year <span class="required">*</span>
687.      </label>
688.
689.      <input
690.      type="number"
691.      id="passingYear"
692.      placeholder="YYYY"
693.      >
694.
695.      <span class="valid-icon">✓</span>
696.
697.      <div class="error" id="passingYearError"></div>
698.
```



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```
699.     </div>
700.
701.     <div class="form-group">
702.
703.         <label>
704.             Percentage / CGPA <span class="required">*</span>
705.         </label>
706.
707.         <input
708.             type="number"
709.             id="percentage"
710.             step="0.01"
711.             placeholder="Enter percentage"
712.         >
713.
714.         <span class="valid-icon">✓</span>
715.
716.     <div class="error" id="percentageError"></div>
717.
718. </div>
719.
720. </div>
721. </div>
722.
723. <!-- ===== -->
724. <!-- PROFESSIONAL -->
725. <!-- ===== -->
726.
727. <div class="section">
728.
729.     <div class="section-heading">
730.
731.     <div class="number">03</div>
732.
733.     <div>
734.     <h2>Professional Information</h2>
735.     <p>Your work experience</p>
736.     </div>
737.
738. </div>
739.
740. <div class="form-grid">
741.
742.     <div class="form-group">
743.
```



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```
744.     <label>
745.     Applying For / Job Position <span class="required">*</span>
746.     </label>
747.
748.     <input
749.     type="text"
750.     id="jobPosition"
751.     placeholder="e.g. Software Developer"
752.     >
753.
754.     <span class="valid-icon">✓</span>
755.
756.     <div class="error" id="jobPositionError"></div>
757.
758.     </div>
759.
760.     <div class="form-group">
761.
762.     <label>
763.     Total Work Experience <span class="required">*</span>
764.     </label>
765.
766.     <select id="experience">
767.
768.     <option value="">
769.     Select experience
770.     </option>
771.
772.     <option>Fresher</option>
773.     <option>0 - 1 Year</option>
774.     <option>1 - 3 Years</option>
775.     <option>3 - 5 Years</option>
776.     <option>More than 5 Years</option>
777.
778.     </select>
779.
780.     <div class="error" id="experienceError"></div>
781.
782.     </div>
783.
784.     <div class="form-group">
785.
786.     <label>
787.     Current / Previous Organization
788.     </label>
789.
```



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```
790.     <input
791.     type="text"
792.     id="organization"
793.     placeholder="Company name"
794.     >
795.
796.     </div>
797.
798.     <div class="form-group">
799.
800.     <label>
801.     Current Designation
802.     </label>
803.
804.     <input
805.     type="text"
806.     id="designation"
807.     placeholder="Your designation"
808.     >
809.
810.     </div>
811.
812.     <div class="form-group">
813.
814.     <label>
815.     Expected Salary <span class="required">*</span>
816.     </label>
817.
818.     <input
819.     type="number"
820.     id="salary"
821.     placeholder="Expected annual salary"
822.     >
823.
824.     <span class="valid-icon">✓</span>
825.
826.     <div class="error" id="salaryError"></div>
827.
828.     </div>
829.
830. </div>
831. </div>
832.
833. <!-- ===== -->
834. <!-- TECHNICAL SKILLS -->
835. <!-- ===== -->
```



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```
836.
837.     <div class="section">
838.
839.     <div class="section-heading">
840.
841.     <div class="number">04</div>
842.
843.     <div>
844.     <h2>Technical Skills</h2>
845.     <p>Showcase your technical knowledge</p>
846.     </div>
847.
848.     </div>
849.
850.     <div class="form-grid">
851.
852.     <div class="form-group full">
853.
854.     <label>
855.     Programming Languages <span class="required">*</span>
856.     </label>
857.
858.     <input
859.     type="text"
860.     id="programming"
861.     placeholder="e.g. C, C++, Java, Python, JavaScript"
862.     >
863.
864.     <div class="skill-box">
865.
866.     <span class="skill">Python</span>
867.     <span class="skill">Java</span>
868.     <span class="skill">C++</span>
869.     <span class="skill">JavaScript</span>
870.
871.     </div>
872.
873.     <div class="error" id="programmingError"></div>
874.
875.     </div>
876.
877.     <div class="form-group full">
878.
879.     <label>
880.     Technical Skills <span class="required">*</span>
881.     </label>
```



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```
882.
883.     <textarea
884.     id="technicalSkills"
885.     placeholder="e.g. HTML, CSS, React, SQL, Git, Machine Learning"
886.     ></textarea>
887.
888.     <div class="error" id="technicalSkillsError"></div>
889.
890. </div>
891.
892. <div class="form-group full">
893.
894.     <label>
895.     Certifications
896.     </label>
897.
898.     <textarea
899.     id="certifications"
900.     placeholder="Enter your certifications"
901.     ></textarea>
902.
903. </div>
904.
905. </div>
906. </div>
907.
908. <!-- ===== -->
909. <!-- ADDITIONAL -->
910. <!-- ===== -->
911.
912. <div class="section">
913.
914.     <div class="section-heading">
915.
916.     <div class="number">05</div>
917.
918.     <div>
919.     <h2>Additional Information</h2>
920.     <p>Anything else you'd like us to know</p>
921.     </div>
922.
923. </div>
924.
925. <div class="form-group">
926.
927.     <label>
```



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```
928.     Cover Letter / Additional Information
929.     </label>
930.
931.     <textarea
932.     id="coverLetter"
933.     placeholder="Write something about yourself..."
934.     ></textarea>
935.
936. </div>
937.
938. </div>
939.
940. <!-- BUTTONS -->
941.
942. <div class="buttons">
943.
944.     <button
945.     type="reset"
946.     class="reset-btn"
947.     id="resetBtn"
948.     >
949.     Reset
950.     </button>
951.
952.     <button
953.     type="submit"
954.     class="submit-btn"
955.     >
956.     Submit Application →
957.     </button>
958.
959. </div>
960.
961. <div
962.     class="success"
963.     id="successMessage"
964.     >
965.     ✓ Application submitted successfully!
966. </div>
967.
968. </form>
969.
970. </div>
971.
972. <div class="footer">
```



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```
973.     Your information will be used only for the recruitment process.
974.     </div>
975.
976.     </div>
977.
978.     <script>
979.
980.     /* =====
981.     HELPER FUNCTIONS
982.     ===== */
983.
984.     function error(input, errorId, message) {
985.
986.         input.classList.remove("valid");
987.         input.classList.add("invalid");
988.
989.         input.closest(".form-group")
990.         ?.classList.remove("is-valid");
991.
992.         document.getElementById(errorId).textContent = message;
993.
994.         return false;
995.     }
996.
997.     function valid(input, errorId) {
998.
999.         input.classList.remove("invalid");
1000.        input.classList.add("valid");
1001.
1002.        input.closest(".form-group")
1003.        ?.classList.add("is-valid");
1004.
1005.        document.getElementById(errorId).textContent = "";
1006.
1007.        return true;
1008.    }
1009.
1010.    /* =====
1011.    NAME
1012.    ===== */
1013.
1014.    function validateName() {
1015.
1016.        let value = fullName.value.trim();
1017.
1018.        if (value === "")
```



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```
1019.     return error(  
1020.     fullName,  
1021.     "fullNameError",  
1022.     "Please enter your full name."  
1023.     );  
1024.  
1025.     if (value.length < 3)  
1026.     return error(  
1027.     fullName,  
1028.     "fullNameError",  
1029.     "Name must contain at least 3 characters."  
1030.     );  
1031.  
1032.     if (!/^[A-Za-z ]+$/.test(value))  
1033.     return error(  
1034.     fullName,  
1035.     "fullNameError",  
1036.     "Name should contain only letters."  
1037.     );  
1038.  
1039.     return valid(fullName, "fullNameError");  
1040. }  
1041.  
  
1042. /* =====  
1043. DOB  
1044. ===== */  
1045.  
1046. function validateDOB() {  
1047.  
1048.     if (dob.value === "")  
1049.     return error(  
1050.     dob,  
1051.     "dobError",  
1052.     "Please select your date of birth."  
1053.     );  
1054.  
1055.     let date = new Date(dob.value);  
1056.     let today = new Date();  
1057.  
1058.     if (date >= today)  
1059.     return error(  
1060.     dob,  
1061.     "dobError",  
1062.     "Date of birth must be in the past."  
1063.     );  
1064.  
1065.     return valid(dob, "dobError");
```



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```
1066.     }
1067.
1068.     /* =====
1069.     EMAIL
1070.     ===== */
1071.
1072.     function validateEmail() {
1073.
1074.         let value = email.value.trim();
1075.
1076.         if (value === "")
1077.             return error(
1078.                 email,
1079.                 "emailError",
1080.                 "Please enter your email address."
1081.             );
1082.
1083.         let pattern =
1084.             /^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$/;
1085.
1086.         if (!pattern.test(value))
1087.             return error(
1088.                 email,
1089.                 "emailError",
1090.                 "Please enter a valid email address."
1091.             );
1092.
1093.         return valid(email, "emailError");
1094.     }
1095.
1096.     /* =====
1097.     MOBILE
1098.     ===== */
1099.
1100.     function validateMobile() {
1101.
1102.         let value = mobile.value;
1103.
1104.         if (value === "")
1105.             return error(
1106.                 mobile,
1107.                 "mobileError",
1108.                 "Please enter your mobile number."
1109.             );
1110.
1111.         if (!/^\d+$/.test(value))
```



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```
1112.     return error(
1113.     mobile,
1114.     "mobileError",
1115.     "Only numbers are allowed."
1116.     );
1117.
1118.     if (value.length < 10)
1119.     return error(
1120.     mobile,
1121.     "mobileError",
1122.     `Please enter 10 digits. ${value.length}/10 entered.`
1123.     );
1124.
1125.     if (!/^[6-9]/.test(value))
1126.     return error(
1127.     mobile,
1128.     "mobileError",
1129.     "Mobile number should start with 6, 7, 8 or 9."
1130.     );
1131.
1132.     return valid(mobile, "mobileError");
1133. }
1134.
1135. /* =====
1136. ADDRESS
1137. ===== */
1138.
1139. function validateAddress() {
1140.
1141.     if (address.value.trim() === "")
1142.     return error(
1143.     address,
1144.     "addressError",
1145.     "Please enter your address."
1146.     );
1147.
1148.     return valid(address, "addressError");
1149. }
1150.
1151. /* =====
1152. CITY
1153. ===== */
1154.
1155. function validateCity() {
1156.
1157.     let value = city.value.trim();
```



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```
1158.
1159.     if (value === "")
1160.         return error(
1161.             city,
1162.             "cityError",
1163.             "Please enter your city."
1164.         );
1165.
1166.     if (!/^[A-Za-z ]+$/.test(value))
1167.         return error(
1168.             city,
1169.             "cityError",
1170.             "City should contain only letters."
1171.         );
1172.
1173.     return valid(city, "cityError");
1174. }
1175.
1176. /* =====
1177. STATE
1178. ===== */
1179.
1180. function validateState() {
1181.
1182.     let value = state.value.trim();
1183.
1184.     if (value === "")
1185.         return error(
1186.             state,
1187.             "stateError",
1188.             "Please enter your state."
1189.         );
1190.
1191.     if (!/^[A-Za-z ]+$/.test(value))
1192.         return error(
1193.             state,
1194.             "stateError",
1195.             "State should contain only letters."
1196.         );
1197.
1198.     return valid(state, "stateError");
1199. }
1200.
1201. /* =====
1202. PIN
1203. ===== */
```



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```
1204.
1205.     function validatePin() {
1206.
1207.         let value = pin.value;
1208.
1209.         if (value === "")
1210.             return error(
1211.                 pin,
1212.                 "pinError",
1213.                 "Please enter your PIN code."
1214.             );
1215.
1216.         if (value.length < 6)
1217.             return error(
1218.                 pin,
1219.                 "pinError",
1220.                 `Please enter 6 digits. ${value.length}/6 entered.`
1221.             );
1222.
1223.         return valid(pin, "pinError");
1224.     }
1225.
1226.     /* =====
1227.     QUALIFICATION
1228.     ===== */
1229.
1230.     function validateQualification() {
1231.
1232.         if (qualification.value === "")
1233.             return error(
1234.                 qualification,
1235.                 "qualificationError",
1236.                 "Please select your qualification."
1237.             );
1238.
1239.         return valid(
1240.             qualification,
1241.             "qualificationError"
1242.         );
1243.     }
1244.
1245.     /* =====
1246.     UNIVERSITY
1247.     ===== */
1248.
1249.     function validateUniversity() {
```



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```
1250.
1251.     if (university.value.trim() === "")
1252.         return error(
1253.             university,
1254.             "universityError",
1255.             "Please enter your university/institute."
1256.         );
1257.
1258.     return valid(
1259.         university,
1260.         "universityError"
1261.     );
1262. }
1263.
1264. /* =====
1265. PASSING YEAR
1266. ===== */
1267.
1268. function validatePassingYear() {
1269.
1270.     let value = Number(passingYear.value);
1271.
1272.     let currentYear =
1273.         new Date().getFullYear();
1274.
1275.     if (!passingYear.value)
1276.         return error(
1277.             passingYear,
1278.             "passingYearError",
1279.             "Please enter passing year."
1280.         );
1281.
1282.     if (value < 1950 || value > currentYear)
1283.         return error(
1284.             passingYear,
1285.             "passingYearError",
1286.             `Enter a year between 1950 and ${currentYear}.`
1287.         );
1288.
1289.     return valid(
1290.         passingYear,
1291.         "passingYearError"
1292.     );
1293. }
1294.
1295. /* =====
```



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```
1296.  PERCENTAGE
1297.  ===== */
1298.
1299.  function validatePercentage() {
1300.
1301.  let value = Number(percentage.value);
1302.
1303.  if (percentage.value === "")
1304.  return error(
1305.  percentage,
1306.  "percentageError",
1307.  "Please enter percentage/CGPA."
1308.  );
1309.
1310.  if (value < 0 || value > 100)
1311.  return error(
1312.  percentage,
1313.  "percentageError",
1314.  "Value must be between 0 and 100."
1315.  );
1316.
1317.  return valid(
1318.  percentage,
1319.  "percentageError"
1320.  );
1321.  }
1322.
1323.  /* =====
1324.  JOB
1325.  ===== */
1326.
1327.  function validateJob() {
1328.
1329.  if (jobPosition.value.trim() === "")
1330.  return error(
1331.  jobPosition,
1332.  "jobPositionError",
1333.  "Please enter the job position."
1334.  );
1335.
1336.  return valid(
1337.  jobPosition,
1338.  "jobPositionError"
1339.  );
1340.  }
1341.
```



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```
1342.  /* =====
1343.  EXPERIENCE
1344.  ===== */
1345.
1346.  function validateExperience() {
1347.
1348.    if (experience.value === "")
1349.      return error(
1350.        experience,
1351.        "experienceError",
1352.        "Please select your experience."
1353.      );
1354.
1355.    return valid(
1356.      experience,
1357.      "experienceError"
1358.    );
1359.  }
1360.
1361.  /* =====
1362.  SALARY
1363.  ===== */
1364.
1365.  function validateSalary() {
1366.
1367.    if (salary.value === "")
1368.      return error(
1369.        salary,
1370.        "salaryError",
1371.        "Please enter expected salary."
1372.      );
1373.
1374.    if (Number(salary.value) < 0)
1375.      return error(
1376.        salary,
1377.        "salaryError",
1378.        "Salary cannot be negative."
1379.      );
1380.
1381.    return valid(
1382.      salary,
1383.      "salaryError"
1384.    );
1385.  }
1386.
1387.  /* =====
```



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```
1388. PROGRAMMING
1389. ===== */
1390.
1391. function validateProgramming() {
1392.
1393.     if (programming.value.trim() === "")
1394.         return error(
1395.             programming,
1396.             "programmingError",
1397.             "Please enter programming languages."
1398.         );
1399.
1400.     return valid(
1401.         programming,
1402.         "programmingError"
1403.     );
1404. }
1405.
1406. /* =====
1407. TECHNICAL SKILLS
1408. ===== */
1409.
1410. function validateTechnical() {
1411.
1412.     if (technicalSkills.value.trim() === "")
1413.         return error(
1414.             technicalSkills,
1415.             "technicalSkillsError",
1416.             "Please enter your technical skills."
1417.         );
1418.
1419.     return valid(
1420.         technicalSkills,
1421.         "technicalSkillsError"
1422.     );
1423. }
1424.
1425. /* =====
1426. GET ELEMENTS
1427. ===== */
1428.
1429. const fullName =
1430.     document.getElementById("fullName");
1431.
1432. const dob =
1433.     document.getElementById("dob");
```



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```
1434.
1435.     const email =
1436.     document.getElementById("email");
1437.
1438.     const mobile =
1439.     document.getElementById("mobile");
1440.
1441.     const address =
1442.     document.getElementById("address");
1443.
1444.     const city =
1445.     document.getElementById("city");
1446.
1447.     const state =
1448.     document.getElementById("state");
1449.
1450.     const pin =
1451.     document.getElementById("pin");
1452.
1453.     const qualification =
1454.     document.getElementById("qualification");
1455.
1456.     const university =
1457.     document.getElementById("university");
1458.
1459.     const passingYear =
1460.     document.getElementById("passingYear");
1461.
1462.     const percentage =
1463.     document.getElementById("percentage");
1464.
1465.     const jobPosition =
1466.     document.getElementById("jobPosition");
1467.
1468.     const experience =
1469.     document.getElementById("experience");
1470.
1471.     const salary =
1472.     document.getElementById("salary");
1473.
1474.     const programming =
1475.     document.getElementById("programming");
1476.
1477.     const technicalSkills =
1478.     document.getElementById("technicalSkills");
1479.
1480.     const form =
1481.     document.getElementById("jobForm");
```



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```
1482.
1483.  /* =====
1484.  REAL-TIME EVENTS
1485.  ===== */
1486.
1487.  fullName.addEventListener(
1488.    "input",
1489.    validateName
1490.  );
1491.
1492.  dob.addEventListener(
1493.    "change",
1494.    validateDOB
1495.  );
1496.
1497.  email.addEventListener(
1498.    "input",
1499.    validateEmail
1500.  );
1501.
1502.  mobile.addEventListener(
1503.    "input",
1504.    function() {
1505.
1506.      mobile.value =
1507.        mobile.value.replace(/\D/g, "");
1508.
1509.      validateMobile();
1510.    }
1511.  );
1512.
1513.  address.addEventListener(
1514.    "input",
1515.    validateAddress
1516.  );
1517.
1518.  city.addEventListener(
1519.    "input",
1520.    validateCity
1521.  );
1522.
1523.  state.addEventListener(
1524.    "input",
1525.    validateState
```



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```
1526. );
1527.
1528. pin.addEventListener(
1529. "input",
1530. function() {
1531.
1532. pin.value =
1533. pin.value.replace(/\D/g, "");
1534.
1535. validatePin();
1536. }
1537. );
1538.
1539. qualification.addEventListener(
1540. "change",
1541. validateQualification
1542. );
1543.
1544. university.addEventListener(
1545. "input",
1546. validateUniversity
1547. );
1548.
1549. passingYear.addEventListener(
1550. "input",
1551. validatePassingYear
1552. );
1553.
1554. percentage.addEventListener(
1555. "input",
1556. validatePercentage
1557. );
1558.
1559. jobPosition.addEventListener(
1560. "input",
1561. validateJob
1562. );
1563.
1564. experience.addEventListener(
1565. "change",
1566. validateExperience
1567. );
```



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```
1568.

1569.     salary.addEventListener(
1570.         "input",
1571.         validateSalary
1572.     );
1573.

1574.     programming.addEventListener(
1575.         "input",
1576.         validateProgramming
1577.     );
1578.

1579.     technicalSkills.addEventListener(
1580.         "input",
1581.         validateTechnical
1582.     );
1583.

1584.     /* =====
1585.     GENDER
1586.     ===== */
1587.
1588.     document.querySelectorAll(
1589.         'input[name="gender"]'
1590.     ).forEach(function (radio) {
1591.
1592.         radio.addEventListener(
1593.             "change",
1594.             function () {
1595.
1596.                 document.getElementById(
1597.                     "genderError"
1598.                 ).textContent = "";
1599.
1600.             }
1601.         );
1602.
1603.     });
1604.

1605.     /* =====
1606.     SUBMIT
1607.     ===== */
1608.

1609.     form.addEventListener(
1610.         "submit",
```



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```
1611.     function(event) {
1612.
1613.         event.preventDefault();
1614.
1615.         let gender =
1616.             document.querySelector(
1617.                 'input[name="gender"]:checked'
1618.             );
1619.
1620.         let genderValid = true;
1621.
1622.         if (!gender) {
1623.
1624.             document.getElementById(
1625.                 "genderError"
1626.             ).textContent =
1627.                 "Please select your gender.";
1628.
1629.             genderValid = false;
1630.         }
1631.
1632.         let allValid =
1633.             validateName() &&
1634.             validateDOB() &&
1635.             genderValid &&
1636.             validateEmail() &&
1637.             validateMobile() &&
1638.             validateAddress() &&
1639.             validateCity() &&
1640.             validateState() &&
1641.             validatePin() &&
1642.             validateQualification() &&
1643.             validateUniversity() &&
1644.             validatePassingYear() &&
1645.             validatePercentage() &&
1646.             validateJob() &&
1647.             validateExperience() &&
1648.             validateSalary() &&
1649.             validateProgramming() &&
1650.             validateTechnical();
1651.
1652.         if (allValid) {
1653.
1654.             document.getElementById(
1655.                 "successMessage"
1656.             ).style.display = "block";
```



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```
1657.
1658.     document.getElementById(
1659.         "successMessage"
1660.     ).scrollIntoView({
1661.         behavior: "smooth"
1662.     });
1663.
1664.     } else {
1665.
1666.         document.getElementById(
1667.             "successMessage"
1668.         ).style.display = "none";
1669.
1670.         let firstInvalid =
1671.         document.querySelector(
1672.             ".invalid"
1673.         );
1674.
1675.         if (firstInvalid) {
1676.
1677.             firstInvalid.scrollIntoView({
1678.                 behavior: "smooth",
1679.                 block: "center"
1680.             });
1681.
1682.             firstInvalid.focus();
1683.
1684.         }
1685.     }
1686.
1687. }
1688. );
1689.
1690. /* =====
1691. RESET
1692. ===== */
1693.
1694. document.getElementById(
1695.     "resetBtn"
1696. ).addEventListener(
1697.     "click",
1698.     function() {
1699.
1700.         setTimeout(function() {
1701.
1702.             document
1703.                 .querySelectorAll(
```



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```
1704.     "input, select, textarea"
1705.     )
1706.     .forEach(function(element) {
1707.
1708.         element.classList.remove(
1709.             "valid",
1710.             "invalid"
1711.         );
1712.
1713.     });
1714.
1715.     document
1716.     .querySelectorAll(
1717.         ".form-group"
1718.     )
1719.     .forEach(function(group) {
1720.
1721.         group.classList.remove(
1722.             "is-valid"
1723.         );
1724.
1725.     });
1726.
1727.     document
1728.     .querySelectorAll(
1729.         ".error"
1730.     )
1731.     .forEach(function(error) {
1732.
1733.         error.textContent = "";
1734.
1735.     });
1736.
1737.     document.getElementById(
1738.         "successMessage"
1739.     ).style.display = "none";
1740.
1741.     }, 10);
1742.
1743.     }
1744.     );
1745.
1746. </script>
1747.
1748. </body>
1749. </html>
```



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4.Output:

127.0.0.1:5500/Exp8/Case_Study8_Aaryan_Kawale_1162.html

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Take the next step in your career. Fill in your details below.

Online Application

01

Personal Information

Tell us about yourself

Full Name *

Aaryan Kawale

✓

Date of Birth *

12/08/2006

Gender *

Male

Female

Other

Email Address *

aaryankawale18@gmail.com

✓

Mobile Number *

6544549846

✓

Address *

Ladikar layout manewada road

✓

City *

nagpur

✓

State *

maharashtra

✓

PIN Code *

440024

✓



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127.0.0.1:5500/Exp8/Case_Study8_Aaryan_Kawale_1162.html

All Bookmarks

Job Application

Take the next step in your career. Fill in your details below.

Online Application

01

Personal Information

Tell us about yourself

Full Name *

Aaryan Kawale

Date of Birth *

12/08/2006

Gender *

Male

Female

Other

Email Address *

aaryankawale18gmail.com

Please enter a valid email address.

Mobile Number *

4544549846

Mobile number should start with 6, 7, 8 or 9.

Address *

Ladikar layout manewada road

City *

nagpur

State *

maharashtra

PIN Code *

44002

Please enter 6 digits. 5/6 entered.



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Experiment No. 9

1. Experiment Title: Implement Local Storage and Session Storage to save user preferences like theme settings.

2. Software/Tools Required: VS Code, Html/JS Compiler...

3. Experiment Program Code:

```
4. <!DOCTYPE html>
5. <html lang="en">
6. <head>
7.     <meta charset="UTF-8">
8.     <meta name="viewport" content="width=device-width, initial-
    scale=1.0">
9.     <title>Theme Preference Manager</title>
10.    <style>
11.        /* CSS Variables for Dynamic Themes */
12.        :root {
13.            --bg-color: #f4f4f9;
14.            --text-color: #333333;
15.            --card-bg: #ffffff;
16.            --border-color: #cccccc;
17.            --btn-primary: #007bff;
18.            --btn-primary-hover: #0056b3;
19.            --btn-danger: #dc3545;
20.            --btn-danger-hover: #bd2130;
21.        }
22.
23.        body.dark-theme {
24.            --bg-color: #121212;
25.            --text-color: #e0e0e0;
26.            --card-bg: #1e1e1e;
27.            --border-color: #444444;
28.            --btn-primary: #bb86fc;
29.            --btn-primary-hover: #3700b3;
30.            --btn-danger: #cf6679;
31.            --btn-danger-hover: #b00020;
32.        }
33.
34.        body {
35.            background-color: var(--bg-color);
36.            color: var(--text-color);
37.            font-family: Arial, sans-serif;
38.            display: flex;
39.            justify-content: center;
40.            align-items: center;
```



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```
41.         height: 100vh;
42.         margin: 0;
43.         transition: background-color 0.3s ease, color 0.3s ease;
44.     }
45.
46.     .container {
47.         background-color: var(--card-bg);
48.         padding: 30px;
49.         border-radius: 10px;
50.         box-shadow: 0 4px 10px rgba(0, 0, 0, 0.2);
51.         width: 350px;
52.         text-align: center;
53.         border: 1px solid var(--border-color);
54.         transition: background-color 0.3s ease, border-color 0.3s
ease;
55.     }
56.
57.     h2 {
58.         margin-bottom: 20px;
59.     }
60.
61.     .control-group {
62.         margin-bottom: 20px;
63.         text-align: left;
64.     }
65.
66.     label {
67.         display: block;
68.         margin-bottom: 8px;
69.         font-weight: bold;
70.     }
71.
72.     select {
73.         width: 100%;
74.         padding: 10px;
75.         border-radius: 5px;
76.         border: 1px solid var(--border-color);
77.         background-color: var(--bg-color);
78.         color: var(--text-color);
79.         font-size: 16px;
80.         cursor: pointer;
81.     }
82.
83.     .button-group {
84.         display: flex;
85.         flex-direction: column;
86.         gap: 10px;
87.     }
```



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```
88.
89.     button {
90.         padding: 10px;
91.         border: none;
92.         border-radius: 5px;
93.         font-size: 15px;
94.         cursor: pointer;
95.         color: #ffffff;
96.         transition: background-color 0.2s ease;
97.     }
98.
99.     .btn-local {
100.         background-color: var(--btn-primary);
101.     }
102.     .btn-local:hover {
103.         background-color: var(--btn-primary-hover);
104.     }
105.
106.     .btn-session {
107.         background-color: #17a2b8;
108.     }
109.     .btn-session:hover {
110.         background-color: #138496;
111.     }
112.
113.     .btn-clear {
114.         background-color: var(--btn-danger);
115.     }
116.     .btn-clear:hover {
117.         background-color: var(--btn-danger-hover);
118.     }
119.
120.     #status-msg {
121.         margin-top: 15px;
122.         font-size: 14px;
123.         font-style: italic;
124.     }
125. </style>
126. </head>
127. <body>
128.
129.     <div class="container">
130.         <h2>Theme Preferences</h2>
131.
132.         <!-- 1. List to select the theme -->
133.         <div class="control-group">
134.             <label for="themeSelect">Select Theme:</label>
135.             <select id="themeSelect">
```



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```
136.         <option value="light">Light Theme</option>
137.         <option value="dark">Dark Theme</option>
138.     </select>
139. </div>
140.
141.     <!-- 2 & 3. Buttons for Storage options and Clearing
142.     preferences -->
143.     <div class="button-group">
144.         <button id="storeLocalBtn" class="btn-local">Store
145.         using LocalStorage</button>
146.         <button id="storeSessionBtn" class="btn-
147.         session">Store using SessionStorage</button>
148.         <button id="clearBtn" class="btn-clear">Clear
149.         Preference</button>
150.     </div>
151.
152.     <div id="status-msg"></div>
153. </div>
154.
155. <script>
156.     const themeSelect =
157.     document.getElementById('themeSelect');
158.     const storeLocalBtn =
159.     document.getElementById('storeLocalBtn');
160.     const storeSessionBtn =
161.     document.getElementById('storeSessionBtn');
162.     const clearBtn = document.getElementById('clearBtn');
163.     const statusMsg = document.getElementById('status-
164.     msg');
165.
166.     // Function to apply the theme visually
167.     function applyTheme(theme) {
168.         if (theme === 'dark') {
169.             document.body.classList.add('dark-theme');
170.             themeSelect.value = 'dark';
171.         } else {
172.             document.body.classList.remove('dark-theme');
173.             themeSelect.value = 'light';
174.         }
175.     }
176.
177.     // Check saved preferences on page load
178.     // Priority: LocalStorage -> SessionStorage -> Default
179.     (Light)
180.     window.addEventListener('DOMContentLoaded', () => {
181.         const savedLocalTheme =
182.         localStorage.getItem('theme_preference');
```



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```
173.         const savedSessionTheme =
    sessionStorage.getItem('theme_preference');
174.
175.         if (savedLocalTheme) {
176.             applyTheme(savedLocalTheme);
177.             statusMsg.textContent = "Loaded from
    LocalStorage.";
178.         } else if (savedSessionTheme) {
179.             applyTheme(savedSessionTheme);
180.             statusMsg.textContent = "Loaded from
    SessionStorage.";
181.         } else {
182.             applyTheme('light'); // Default
183.         }
184.     });
185.
186.     // Live preview when changing the dropdown option
187.     themeSelect.addEventListener('change', (e) => {
188.         applyTheme(e.target.value);
189.         statusMsg.textContent = "Theme preview changed (Not
    saved yet).";
190.     });
191.
192.     // Store preference in LocalStorage (Persists even
    after closing and reopening the browser/tab)
193.     storeLocalBtn.addEventListener('click', () => {
194.         const selectedTheme = themeSelect.value;
195.         localStorage.setItem('theme_preference',
    selectedTheme);
196.         sessionStorage.removeItem('theme_preference'); //
    Clear session to avoid conflicts
197.         statusMsg.textContent = "Saved to LocalStorage!
    (Persists across browser sessions)";
198.     });
199.
200.     // Store preference in SessionStorage (Persists on
    refresh, but resets when tab/window is closed)
201.     storeSessionBtn.addEventListener('click', () => {
202.         const selectedTheme = themeSelect.value;
203.         sessionStorage.setItem('theme_preference',
    selectedTheme);
204.         localStorage.removeItem('theme_preference'); //
    Clear local to avoid conflicts
205.         statusMsg.textContent = "Saved to SessionStorage!
    (Resets when tab/window closes)";
206.     });
207.
208.     // Clear preferences
```



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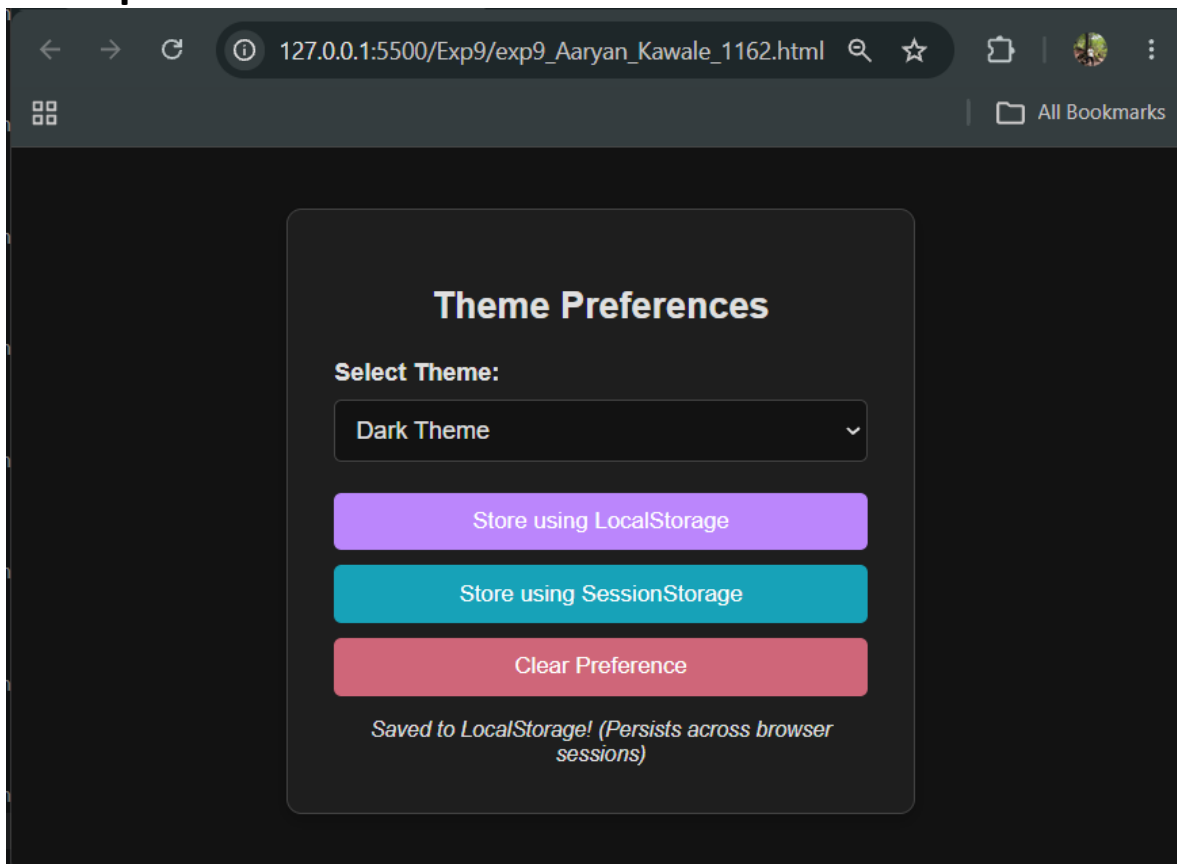
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```
209.         clearBtn.addEventListener('click', () => {
210.             localStorage.removeItem('theme_preference');
211.             sessionStorage.removeItem('theme_preference');
212.             applyTheme('light'); // Reset to default
213.             statusMsg.textContent = "Preferences cleared. Reset
to default.";
214.         });
215.     </script>
216.
217. </body>
218. </html>
```

4.Output:





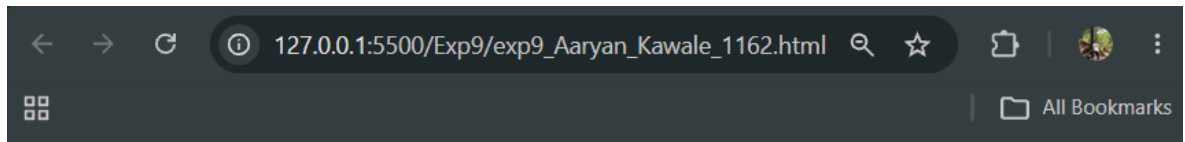
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Theme Preferences

Select Theme:

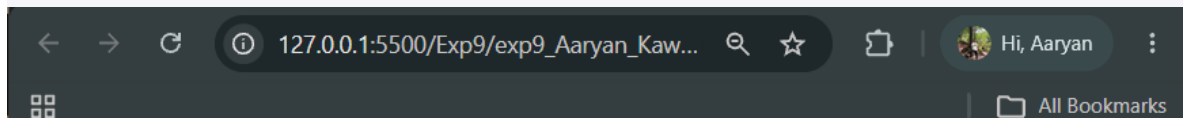
Light Theme

Store using LocalStorage

Store using SessionStorage

Clear Preference

Saved to SessionStorage! (Resets when tab/window closes)



Theme Preferences

Select Theme:

Dark Theme

Store using LocalStorage

Store using SessionStorage

Clear Preference

Loaded from LocalStorage.



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Experiment No. 10

- 1. Experiment Title:** Load and display JSON data using fetch() and Query\$.getJSON(); render result dynamically in a table.
- 2. Software/Tools Required:** VS Code, Html/JS Compiler...
- 3. Experiment Program Code:**

```
4. <!DOCTYPE html>
5. <html lang="en">
6. <head>
7.     <meta charset="UTF-8">
8.     <meta name="viewport" content="width=device-width, initial-
    scale=1.0">
9.     <title>Local JSON Data Table Demo</title>
10.
11.     <!-- Load jQuery from CDN -->
12.     <script src="https://code.jquery.com/jquery-
    3.7.1.min.js"></script>
13.
14.     <style>
15.         /* --- CSS STYLES --- */
16.         body {
17.             font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-
    serif;
18.             background-color: #f4f4f9;
19.             padding: 20px;
20.             color: #333;
21.         }
22.
23.         .container {
24.             max-width: 900px;
25.             margin: 0 auto;
26.             background: white;
27.             padding: 20px;
28.             border-radius: 8px;
29.             box-shadow: 0 2px 10px rgba(0,0,0,0.1);
30.         }
31.
32.         h1 {
33.             text-align: center;
34.             color: #2c3e50;
35.         }
36.
37.         .controls {
38.             display: flex;
```



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```
39.         justify-content: center;
40.         gap: 15px;
41.         margin-bottom: 20px;
42.     }
43.
44.     button {
45.         padding: 10px 20px;
46.         font-size: 16px;
47.         cursor: pointer;
48.         border: none;
49.         border-radius: 5px;
50.         transition: background 0.3s;
51.     }
52.
53.     .btn-fetch {
54.         background-color: #3498db;
55.         color: white;
56.     }
57.     .btn-fetch:hover { background-color: #2980b9; }
58.
59.     .btn-jquery {
60.         background-color: #e67e22;
61.         color: white;
62.     }
63.     .btn-jquery:hover { background-color: #d35400; }
64.
65.     .status-msg {
66.         text-align: center;
67.         margin-bottom: 10px;
68.         font-style: italic;
69.         height: 20px;
70.         font-weight: 500;
71.     }
72.
73.     /* Table Styles */
74.     table {
75.         width: 100%;
76.         border-collapse: collapse;
77.         margin-top: 10px;
78.     }
79.
80.     thead {
81.         background-color: #2c3e50;
82.         color: white;
83.     }
84.
85.     th, td {
86.         padding: 12px 15px;
```



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```
87.         text-align: left;
88.         border-bottom: 1px solid #ddd;
89.     }
90.
91.     tbody tr:hover {
92.         background-color: #f1f1f1;
93.     }
94.
95.     /* Loading Spinner */
96.     .loader {
97.         border: 4px solid #f3f3f3;
98.         border-top: 4px solid #3498db;
99.         border-radius: 50%;
100.         width: 30px;
101.         height: 30px;
102.         animation: spin 1s linear infinite;
103.         margin: 20px auto;
104.         display: none; /* Hidden by default */
105.     }
106.
107.     @keyframes spin {
108.         0% { transform: rotate(0deg); }
109.         100% { transform: rotate(360deg); }
110.     }
111. </style>
112. </head>
113. <body>
114.
115.     <div class="container">
116.         <h1>Team Members</h1>
117.
118.         <div class="controls">
119.             <button class="btn-fetch"
120.                 onclick="loadWithFetch()">Load via Fetch Logic</button>
121.             <button class="btn-jquery"
122.                 onclick="loadWithjQuery()">Load via jQuery Logic</button>
123.         </div>
124.
125.         <div id="status" class="status-msg">Click a button to load
126.             team data.</div>
127.         <div id="loader" class="loader"></div>
128.
129.         <table id="userTable">
130.             <thead>
131.                 <tr>
```



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```
132.         <th>Role</th>
133.     </tr>
134. </thead>
135. <tbody>
136.     <!-- Rows will be injected here dynamically -->
137. </tbody>
138. </table>
139. </div>
140.
141. <script>
142.     // --- JAVASCRIPT LOGIC ---
143.
144.     const localTeamData = [
145.         {
146.             "id": 24070521162,
147.             "name": "Aaryan Kawale",
148.             "email":
149.                 "aaryan.kawale.batch2024@sitnagpur.siu.edu.in",
150.             "role": "HR"
151.         },
152.         {
153.             "id": 24070521146,
154.             "name": "Soumya",
155.             "email":
156.                 "soumya.bajirao.batch2024@sitnagpur.siu.edu.in",
157.             "role": "Frontend Developer"
158.         },
159.         {
160.             "id": 24070521211,
161.             "name": "Janhavi",
162.             "email":
163.                 "janhavi.admane.batch2024@sitnagpur.siu.edu.in",
164.             "role": "UI/UX Designer"
165.         },
166.         {
167.             "id": 24070521143,
168.             "name": "Arman",
169.             "email":
170.                 "arman.meshram.batch2024@sitnagpur.siu.edu.in",
171.             "role": "Backend Engineer"
172.         },
173.         {
174.             "id": 24070521144,
175.             "name": "Atharva",
176.             "email":
177.                 "atharva.suryawanshi.batch2024@sitnagpur.siu.edu.in",
178.             "role": "Project Manager"
179.         }
180.     ]
181. }
```



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```
175.     ];
176.
177.     const tableBody = document.querySelector('#userTable
tbody');
178.     const statusMsg = document.getElementById('status');
179.     const loader = document.getElementById('loader');
180.
181.     // Helper: Show/Hide Loader with optional custom completion
text
182.     function toggleLoader(show, successText = "Data loaded
successfully.") {
183.         loader.style.display = show ? 'block' : 'none';
184.         if (show) {
185.             statusMsg.textContent = "Processing data...";
186.             tableBody.innerHTML = ""; // Clear existing data
187.         } else {
188.             statusMsg.textContent = successText;
189.         }
190.     }
191.
192.     // Helper: Generate HTML Row String
193.     function createRow(user) {
194.         return `
195.             <tr>
196.                 <td>${user.id}</td>
197.                 <td><strong>${user.name}</strong></td>
198.                 <td>${user.email}</td>
199.                 <td><span style="background:#e1f5fe; padding:
4px 8px; border-radius: 4px; color: #0277bd;">${user.role}</span></td>
200.             </tr>
201.         `;
202.     }
203.
204.     /* -----
-
205.     METHOD 1: Simulating Fetch API with Local Data
206.     -----
- */
207.     function loadWithFetch() {
208.         toggleLoader(true);
209.
210.         new Promise((resolve) => {
211.             setTimeout(() => resolve(localTeamData), 800);
212.         })
213.         .then(data => {
214.             let htmlContent = '';
215.             data.forEach(user => {
216.                 htmlContent += createRow(user);
```



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```
217.         });
218.
219.         tableBody.innerHTML = htmlContent;
220.         toggleLoader(false, "Data loaded via Fetch
logic."); // Passing specific message
221.     })
222.     .catch(error => {
223.         console.error('Error:', error);
224.         statusMsg.textContent = "Error processing data.";
225.         toggleLoader(false, "Failed to load data.");
226.     });
227. }
228.
229. /* -----
-
230.     METHOD 2: jQuery Logic with Local Data
231.     -----
- */
232.     function loadWithjQuery() {
233.         toggleLoader(true);
234.
235.         setTimeout(function() {
236.             let htmlContent = '';
237.
238.             $.each(localTeamData, function(index, user) {
239.                 htmlContent += createRow(user);
240.             });
241.
242.             $('#userTable tbody').html(htmlContent);
243.             toggleLoader(false, "Data loaded via jQuery
logic."); // Passing specific message
244.
245.             }, 800);
246.         }
247.     </script>
248.
249. </body>
250. </html>
```



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4.Output:

127.0.0.1:5500/Exp10/exp10_Aaryan_Kawale_1162.html

Team Members

Load via Fetch Logic

Load via jQuery Logic

Click a button to load team data.

ID	Name	Email	Role
----	------	-------	------

127.0.0.1:5500/Exp10/exp10_Aaryan_Kawale_1162.html

Team Members

Load via Fetch Logic

Load via jQuery Logic

Data loaded via jQuery logic.

ID	Name	Email	Role
24070521162	Aaryan Kawale	aaryan.kawale.batch2024@sitnagpur.siu.edu.in	HR
24070521146	Soumya	soumya.bajirao.batch2024@sitnagpur.siu.edu.in	Frontend Developer
24070521211	Janhavi	janhavi.admane.batch2024@sitnagpur.siu.edu.in	UI/UX Designer
24070521143	Arman	arman.meshram.batch2024@sitnagpur.siu.edu.in	Backend Engineer

127.0.0.1:5500/Exp10/exp10_Aaryan_Kawale_1162.html

Team Members

Load via Fetch Logic

Load via jQuery Logic

Data loaded via Fetch logic.

ID	Name	Email	Role
24070521162	Aaryan Kawale	aaryan.kawale.batch2024@sitnagpur.siu.edu.in	HR
24070521146	Soumya	soumya.bajirao.batch2024@sitnagpur.siu.edu.in	Frontend Developer
24070521211	Janhavi	janhavi.admane.batch2024@sitnagpur.siu.edu.in	UI/UX Designer
24070521143	Arman	arman.meshram.batch2024@sitnagpur.siu.edu.in	Backend Engineer